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Cognitive and artificial intelligence evaluation framework

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Abstract
The Cognitive and Artificial Intelligence Evaluation (CAIE) framework provides a structured and domain-independent methodology for assessing the intelligence of artificial and information systems in a broader perspective. The primary achievement of this research is the categorization of over ninety cognitive features into six evaluation zones, supported by a two-stage scoring model that combines detailed feature-level analysis with higher-level structural interpretation. This approach has proven effective in identifying system maturity and developmental potential, offering systematic insights into both strengths and weaknesses across cognitive domains. The practical validation through use-case analysis demonstrates that CAIE is adaptable to diverse technological contexts, enabling consistent comparison between AI and non-AI systems. By treating cognitive features as measurable and comparable attributes, the framework introduces a coherent mechanism for benchmarking, scalability, and strategic development. The main contribution of this work lies in advancing both academic research and real-world implementation by delivering a cognitively informed, practically relevant tool that bridges theoretical evaluation concepts with actionable methods for designing and improving intelligent systems.

Keywords Artificial intelligence · Cognitive intelligence · Cognitive science · Cognitive information systems · Intelligence evaluation · Intelligence scoring systems

1 Introduction

The accelerating integration of Cognitive and Artificial Intelligence capabilities across domains has created an urgent need for rigorous methods to evaluate system intelligence. From Enterprise Architectures to robotics and industrial automation, intelligent systems increasingly influence how decisions are made, problems are solved, and interactions are

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conducted. Assessing their intelligence is therefore essential not only to understand current capabilities but also to guide future development.

Although existing AI-driven platforms demonstrate high levels of operational efficiency and task-specific automation, they typically address isolated needs. This results in a limited capacity to assess overall system intelligence—particularly in terms of adaptability, reasoning, or collaboration. Bridging this gap requires the incorporation of Cognitive Intelligence as a central component. Cognitive systems are not necessarily data-driven, but most modern cognitive systems rely on data-driven methods—where data-driven is a methodological feature, not a conceptual requirement of cognitive systems—to some extent, but must be capable of autonomous learning, adaptation, reasoning, optimization, and collaboration (Wang 2017; Zmigrod et al. 2021). Given the growing diversity of intelligent applications, methodologies, and architectures, there is a pressing need to establish comparability across heterogeneous systems. Without scalable and context-sensitive evaluation frameworks, it remains difficult to determine how effectively a system meets the expectations of its domain. While the subjective assessment of system performance remains valuable, quantitative and structured methods are required to support reliable evaluation and benchmarking.

Current assessment tools often prioritize efficiency or performance in narrow functional areas, such as machine learning or pattern recognition, and rarely account for broader cognitive dimensions such as memory, adaptation, or interaction. Consequently, they provide only a partial view of system intelligence. Several established models—such as Russell and Norvig's AI categorization, Chollet's generalization metric, the SOAR architecture, ACT-R, OpenCog's CogPrime, and IBM Watson's Cognitive Computing Framework—have advanced understanding in discrete areas such as problem-solving, memory modeling, or probabilistic reasoning. However, they often remain limited to theoretical or domain-specific applications and lack mechanisms for cross-domain comparability and user-driven customization.

To address these limitations, a multidimensional, scalable approach is proposed that incorporates both objective performance metrics and user-defined relevance criteria. The aim is not only to assess operational success but also to capture cognitive maturity in terms of how a system learns, adapts, reasons, interacts, and retains information. The Cognitive and Artificial Intelligence Evaluation (CAIE) framework responds to this need by offering a structured taxonomy of cognitive features and a two-stage scoring model, enabling system-agnostic, context-aware intelligence evaluation.

1.1 Research scope

As a consequence, research questions (RQs) arise, which form the basis of the research. One of the research questions (RQ-1) is whether it is possible to compare Cognitive and AI systems in a way that provides an objective measure of the applicability and usability of the system. Closely related to this is the next question (RQ-2), whether (Cognitive and AI systems related) measurability can be ensured in an adaptive way by developing and applying a rating system that takes into account the individual preferences of the user in rating and evaluation. The third question (RQ-3), which also needs to be answered, is whether it is possible to treat the level of cognitive ability of AI-based systems as a key factor in the scoring process, which allows to infer the need for CIS based on the user's needs and preferences.

These research questions form the foundation of the proposed methodology, driving the development of an evaluation framework capable of assessing the 'goodness' of Cognitive and AI systems. By focusing on objectivity, adaptability, and cognitive features, the framework ensures that diverse systems can be measured accurately and compared meaningfully, providing valuable insights for their improvement and future development.

To support the research objectives and proposed framework, the paper is structured as follows: Sect. 2 describes the applied methodology, including both theoretical and empirical components. Section 3 outlines the theoretical background and analyzes existing models. Section 4 introduces the functional zoning of the CAIE framework, followed by Sect. 5, which defines the evaluation levels and maturity logic. Section 6 details the two-stage scoring model and composite metrics. Section 7 presents a practical use-case on SAP BTP. Section 8 reports the results, Sect. 9 discusses the findings and limitations, and finally, Sect. 10 concludes the paper and outlines future research directions.

2 Methodology

This section outlines the methodological foundations of the research, integrating both theoretical exploration and empirical validation to assess the applicability and structure of the CAIE framework.

The methodology is divided into two core components: a systematic literature review based on PRISMA standards, and a quantitative survey-based analysis of the SAP Business Technology Platform (SAP BTP) through the lens of Cognitive and Artificial Intelligence evaluation. SAP (originally Systeme, Anwendungen und Produkte in der Datenverarbeitung—which is Systems, Applications, and Products in Data Processing) is a German multinational software company founded in 1972 and the world's leading provider of Enterprise Resource Planning (ERP) software. SAP BTP was selected as a case environment because it consolidates AI-driven application development, data management, analytics, and integration into a unified enterprise platform. It includes core solutions such as SAP HANA, SAP Analytics Cloud, SAP Integration Suite, and SAP Extension Suite, which collectively represent a AI system capable of adaptive, large-scale enterprise operations (Stoykova and Hrischev 2024; Putnoki 2024). This makes SAP BTP suitable for surveying under the CAIE framework, as it exposes a wide spectrum of AI functionalities—including automation, prediction, integration, and personalization—that directly relate to the Research Questions on system intelligence, decision support, and adaptive evaluation. At the same time, it is emphasized that CAIE is not specific to SAP technologies. The framework is designed to evaluate the maturity and intelligence of any AI-based system, whether enterprise-oriented or domain-specific. The SAP BTP case thus serves as a demonstrative application, providing practical data to illustrate the framework's usability, while the results highlight CAIE's broader relevance across different technological contexts.

The dual-layered design helps ensure that the framework remains grounded in current academic research while also reflecting insights from practical, real-world data. The literature review establishes a conceptual foundation by analyzing recent research trends across six key cognitive domains. The survey, in turn, validates the practicality, usability, and measurable structure of the framework using actual system-level perceptions from BTP users.

By combining evidence from both structured academic analysis and operational feedback, the methodology provides a balanced and multi-perspective validation of the CAIE framework, ensuring real-world relevance.

2.1 PRISMA-based systematic literature review

This study focused on six identified categories of cognitive functions and AI capabilities within intelligent systems. To ensure that the findings reflect the most recent and state-of-the-art developments in these domains, the literature review was limited to the publication years 2020 to 2024.

The identification phase was carried out using five major academic databases: Google Scholar, PubMed, SpringerLink, ScienceDirect, and Web of Science. The cumulative number of keyword-based results retrieved from each database was as follows: Google Scholar (7,342,700), PubMed (211,278), SpringerLink (1,086,271), ScienceDirect (780,315), and Web of Science (111,045).

The analysis of keyword trends from 2020 to 2024 revealed clear shifts in research focus across the six categories. Notably, there was a steady increase in publications related to Interaction and Personalization, indicating growing academic interest in adaptive and user-centered systems. In contrast, a gradual decline was observed in Optimization and Operation Efficiency, suggesting a relative decrease in emphasis on performance-driven solutions during the same period. The annual distribution of selected studies following whole PRISMA methodology is presented in Table 1.

Following the initial retrieval, the dataset was refined to include only publications where the selected keywords appeared in the title. A second-level refinement focused on title-level keyword combinations, yielding a total of 189 records.

In the screening phase, records collected from different databases were first deduplicated, followed by thematic screening based on the six predefined categories. This led to the exclusion of irrelevant or off-topic publications. During the eligibility assessment, full-text access was ensured, and studies were reviewed to verify alignment with the conceptual and methodological focus of the research. This process resulted in the removal of 71 records, leaving 118 studies for final evaluation. As part of the inclusion phase, the remaining 118 studies were subjected to detailed content analysis, which involved reviewing both abstracts and full texts. At this stage, priority was given to publications with robust academic refer-

Table 1 Annual occurrence of keyword categories and selected sources (Final Selected)

Keywords	2020	2021	2022	2023	2024
Learning and Adaptation	446,701	460,250	434,761	380,883	301,316
Perception and Awareness	479,687	448,950	425,220	318,532	267,628
Reasoning and Decision-Making	210,984	205,528	200,293	189,688	161,881
Interaction and Personalization	81,295	87,452	92,997	106,704	137,000
Memory	76,412	80,015	77,309	73,148	62,659
Optimization and Operation Efficiency	892,286	908,525	815,355	585,908	522,242
Final Selected	**14**	**18**	**34**	**26**	**26**

encing and, where possible, those published with the approval of at least two independent reviewers.

2.2 Quantitative evaluation design

To empirically support the research questions and validate the structure of the CAIE framework, a user-oriented quantitative survey was conducted. The objective was to assess how key Cognitive and Artificial Intelligence features are perceived and experienced within SAP Business Technology Platform (SAP BTP).

The questionnaire was structured around the functional categories defined in Sect. 4, covering features from (A) Learning and Adaptation to (F) Optimization and Operational Efficiency. Each feature was evaluated along three axes:

- *Implementation status* a binary indicator capturing whether the feature is present.
- *Perceived performance* a 1–10 rating reflecting the user's experience with the feature in SAP BTP.
- *Importance weight* a 1–10 score representing the perceived importance of the feature from the user's perspective.

These values formed the input for calculating Zoned Unified Composite Score (*ZUCS*) and ultimately generating the System-Zoned Unified Composite Score (*SZUCS*) for enable benchmarking. The survey results provided the foundation for performance-based modeling, grounded in actual user experience. This allowed a structured and scalable evaluation of SAP BTP's cognitive features within the CAIE framework.

3 Background

The study of cognition and intelligence has been shaped by insights from cognitive science, psychology, neuroscience, and artificial intelligence, each contributing distinct perspectives on how intelligence can be explained, reconstructed, and assessed. Over the past decades, numerous frameworks, architectures, and models have been developed to evaluate system intelligence, offering valuable yet fragmented insights into cognitive and computational performance. Nevertheless, a consolidated perspective is essential, one that systematically integrates these contributions and establishes a coherent foundation for capturing the full spectrum of cognitive capabilities.

Cognition is taken as the ensemble of processes—biological or artificial—by which an agent acquires, transforms, stores, retrieves, and applies knowledge to guide action. While artificial cognition applies analogous principles within computational architectures—integrating perception, learning, memory, and reasoning into adaptive loops (Zhang 2019; Sun 2017). Likewise, Cognitive Information Systems (CIS) extend these principles to hybrid Carbon–Silicon environments where the Carbon (Human) agent and Silicon (Machine) agent collaborate through adaptive, context-aware information exchange (Baranyi and Csapo 2024). Intelligence is treated as goal-directed effectiveness across tasks and environments, explicitly tied to generalization, adaptability, and efficiency of learning. In AI theory, intelligence can be formalized as skill-acquisition efficiency over a scope of tasks, relative

to the priors and experience available to the learner. This view integrates with—but is distinct from—psychometric perspectives and universal tests that aim to evaluate agents across diverse environments and time. Artificial Intelligence (AI) is the science and engineering of making intelligent machines—especially intelligent computer programs—capable of performing tasks that, if executed by humans, would require intelligence. This includes both narrow, specialized models trained for specific purposes and more general, adaptive systems (Russell and Norvig 2016; Putnoki and Orosz 2023). In enterprise IT, system-wide intelligence is a data platform where financial, logistical, and customer data are analyzed together, and AI provides integrated insights. In cognitive assessment, system-wide intelligence shows how well the system can coordinate the results of all functional areas into a coherent, high-level operation. Cognitive Intelligence (CI) is a subset of intelligence grounded in higher-order cognitive processes such as reasoning, planning, and transfer. In artificial systems, CI emerges when perception, learning, memory, and reasoning work together and can be applied to new tasks with reconfiguration. Research on cognitive intelligence focuses on embedding human cognitive capabilities into intelligent systems and equipping machines with human-like abilities (Ren et al. 2023). A related but narrower research direction, Cognitive AI (CAI), applies these principles within Artificial Intelligence, explicitly modeling human-like cognitive processes—often combining symbolic reasoning with sub-symbolic learning—while emphasizing context-sensitive adaptation, explainability, and alignment with human cognitive models (Zhao et al. 2022; Sun 2017). Cognitive Systems integrate perception, memory, learning, and reasoning within a unified architecture so that behavior adapts dynamically to changing context. The designation cognitive refers to coordinated, context-aware processes that preserve and update internal state across interactions, which includes Carbon–Silicon collaborations where the agents co-produce decisions via structured, adaptive infocommunication. A system that includes AI might embed one or several models (e.g., a classifier, a recommender) without coordinating them through a shared contextual model or adaptive reasoning loop. In contrast, a cognitive system maintains and applies context across tasks, updates internal representations from feedback, and adapts decision policies over time—often in tandem with human partners via structured infocommunication(Vernon 2014; Mattyasovszky-Philipp et al. 2022).

Cognitive functions are the core mental and information-processing activities that enable humans to perceive, understand, and use acquired knowledge. They are rooted in neurocognitive mechanisms of the brain, supporting intellectual functioning and adaptive behavior. Cognitive functions include the following:

- perception and awareness (e.g., sight, hearing, touch, etc.),
- attention (e.g., maintaining attention, deliberately diverting attention, etc.),
- thinking (e.g., problem solving, planning, decision making, reasoning, calculating, etc.),
- memory (e.g., short-term memory, recalling childhood memories, etc.),
- other skills (e.g., learning, speech, spatial orientation, etc.).

The scientific study of human cognitive functions is the subject of cognitive psychology and cognitive neuroscience (Zhang 2019; Mattyasovszky-Philipp et al. 2022; Anna 2024).

3.1 Existing cognitive architectures and evaluation models

This section distinguishes the literature along two trajectories. The first examines cognitive architectures and models of cognition, which emulate the mechanisms of human thought through unified frameworks that integrate multiple functions. The second considers classification and evaluation approaches, which provide taxonomies and metrics to assess and compare different forms of intelligence, focusing on systematic measurement rather than emulation.

Cognitive architectures are computational frameworks designed to reproduce human-like cognition by integrating processes such as perception, memory, reasoning, learning, and decision-making within a unified structure. As a continuation of the discussion in the previous section, these architectures seek not only to model isolated functions but also to capture their interaction in order to simulate complex, adaptive behavior. A comprehensive review by Kotseruba offers an overview of four decades of research on cognitive architectures, covering 84 distinct architectures developed across 900 practical projects. These projects span diverse domains including robotics, psychological experiments, games and puzzles, categorization and clustering, computer vision, human performance modeling (HPM), human–robot and human–computer interaction (HRI/HCI), virtual agents, and natural language processing (NLP). Cognitive architectures borrow from a wide range of disciplines—extending from psychoanalysis and cognitive psychology to neuroscience and computer science—highlighting their multidisciplinary foundation (Kotseruba and Tsotsos 2020). The review distilled broader cognitive abilities into perception, attention, action selection, motivation, actuation, memory, learning, reasoning, meta-reasoning, social interaction, and creativity. These categories were subsequently consolidated and mapped into the CAIE taxonomy, ensuring alignment with a coherent cognitive function-based structure. However some functions, rather than standing as independent domains, are treated as subfunctions of broader categories. For example, meta-reasoning is considered a feature within the Reasoning and Decision-Making category, while attention is placed within Perception and Awareness. This mapping provides sharper analytical focus by reducing redundancy and highlighting the core cognitive functions as categories. Through this structure, categories (A) Learning and Adaptation, (B) Perception and Awareness, (C) Reasoning and Decision-Making, (D) Interaction and Personalization, (E) Memory establish clear continuity with the universally valid cognitive functions discussed earlier, confirming their role as foundational across architectures. In addition, (F) Optimization and Operation Efficiency extends the taxonomy by capturing a dimension often implicit but rarely formalized in prior reviews. Cognitive science research supports this inclusion, as efficiency, particularly learning efficiency increasingly recognized as an essential cognitive function (Hoffman 2012). These zones correspond to the categories defined in Table 1 and detailed in Sect. 4. This explicit mapping ensures that CAIE's categorization is firmly grounded in human cognition studies while remaining applicable to artificial systems. The following comparative Table 2 illustrates how the most used, common cognitive architectures map onto the defined zones, highlighting their respective strengths, practical applications, and corresponding competency areas (Kotseruba and Tsotsos 2020; Adams et al. 2012).

The comparative mapping highlights that most cognitive architectures cover learning, perception, reasoning, and memory, while interaction and especially optimization remain less developed. ACT-R (Adaptive Control of Thought—Rational), mostly applied in psy-

Table 2 Mapping of cognitive functions across major cognitive architectures

	ACT-R	ART	SOAR	HTM	Leabra	CLARION	RCS	SPA	EPIC	IMPRINT	LIDA	Disciple	CogPrime
A	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes
B	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
C	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
D	Yes	No	Yes	No	No	Yes	No	No	No	No	Yes	Yes	Yes
E	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F	No	No	No	No	No	No	No	No	No	No	No	No	No

chological experiments, is among the most established. It excels at modeling human cognition through procedural and declarative memory, making it a central tool for simulating human-like processes. However, its adaptability and multi-domain intelligence are limited, restricting its broader applicability beyond controlled experiments (Kotseruba and Tsotsos 2020; Anderson 1996). Soar, widely used in categorization, clustering, and virtual agent design, specializes in problem-solving and chunking-based learning by breaking down complex tasks into manageable units. Its cyclical memory processes are valuable, and it has been deployed in robotics, games, and simulations, yet it does not extend well to collaborative processes or ethical reasoning (Kotseruba and Tsotsos 2020; Laird 2019). ART (Adaptive Resonance Theory), primarily used for categorization and clustering, contributes powerful mechanisms for pattern recognition and computer vision but falls short in higher-level reasoning and interactive capabilities. HTM (Hierarchical Temporal Memory), often used in computer vision and anomaly detection, offers biologically inspired sequence learning and prediction, though it remains limited by weak interaction and underdeveloped long-term memory handling. Leabra, grounded in psychological experiments, integrates Hebbian and error-driven learning and has proven effective in perception and attention studies. Despite its neuroscientific fidelity, its application outside experimental domains remains constrained. CLARION, also rooted in psychological experimentation, stands out as a hybrid model balancing implicit and explicit processes, and is one of the few addressing creativity and insight-based problem-solving. However, its complexity reduces scalability in broader applications (Sun 2017; Kotseruba and Tsotsos 2020). Architectures such as RCS and LIDA emphasize robotics and real-time adaptation. RCS is rooted in industrial control, while LIDA integrates attention management, decision-making, and learning, focusing on short-term memory and real-time processing. Both contribute meaningfully to their niches but fall short in optimization and higher-level self-reflection (Kotseruba and Tsotsos 2020; Franklin et al. 2007). EPIC and IMPRINT represent narrower models designed for multitasking and human performance prediction, offering precision but lacking robust learning and adaptability. Disciple, widely applied in human–robot and human–computer interaction (HRI/HCI), is notable for its expert-driven knowledge acquisition and decision support, enabling effective human–system collaboration. However, it struggles with scalability in large, dynamic environments. CogPrime (OpenCog), primarily used in HRI/HCI and natural language processing, aspires to artificial general intelligence by combining probabilistic reasoning, multi-domain adaptability, and knowledge representation through AtomSpace. Despite these ambitions, it lacks robust mechanisms for optimization and long-term feedback integration, limiting its practical utility (Kotseruba and Tsotsos 2020; Goertzel et al. 2014). Similarly, AIS stands out for explicitly addressing creativity and interaction, though its applications remain restricted to limited contexts. Overall, the survey confirms that cognitive architectures such as ACT-R, Soar, CLARION, LIDA, Disciple, AIS, and CogPrime attempt to span most cognitive functions. From a CAIE perspective, these architectures provide valuable insights into modeling cognition but remain incomplete for real-world applications, as they prioritize psychological fidelity or domain-specific functions over scalability, efficiency, and systemic adaptability.

Beyond individual architectures, several theoretical and applied models have extended the scope of cognition research by emphasizing broader principles of reasoning, interaction, and systemic integration. Biologically Inspired Cognitive Architectures (BICA) were developed to balance rational and emotional reasoning, contributing significantly to interac-

tion and reasoning studies. Despite these strengths, BICA remains limited in operational efficiency and cross-domain learning, reducing its utility in applied contexts (Samsonovich 2013). IBM Watson's Cognitive Computing Framework represents a more applied model, integrating symbolic and probabilistic reasoning with natural language processing and adaptive learning. Its strengths lie in learning and reasoning capabilities, particularly in language-driven environments, though it lacks explicit mechanisms for adaptability and self-reflection, constraining its use as a self-improving cognitive system (Khalil et al. 2021). Human-Level AI Systems (HLAI) focus on meta-learning, commonsense reasoning, and adaptability, reinforcing learning and reasoning while aspiring to more generalizable forms of intelligence. However, these systems tend to underrepresent collaboration and resource efficiency, leaving notable gaps in enterprise applications where integration and operational constraints are essential.

From a more theoretical perspective, Wang's Systems Theories provide a general systems-based approach to learning and optimization, expanding the discourse on cognition as a systemic process. Nevertheless, interaction and memory remain underdeveloped within this framework, limiting its comprehensiveness (Wang 2007). In contrast, Philipp's and Baranyi's Cognitive Information Systems are particularly relevant for enterprise environments, emphasizing infocommunication and system integration. These models converge perception, reasoning, and interaction to support organizational decision-making, offering valuable insights into system-level cognition (Molnár and Mattyasovszky-Philipp 2023; Mattyasovszky-Philipp and Molnár 2023; Baranyi et al. 2015; Baranyi and Csapo 2024). Yet, they do not provide explicit mechanisms for evaluating intelligence maturity, indicating how well a system integrates, scales, and coordinates functions across domains to achieve higher-order intelligent behavior. This restricts their ability to function as comprehensive diagnostic frameworks, a gap that the CAIE framework explicitly addresses through staged evaluation and zone-level diagnostics.

In parallel, several approaches aim less at modeling cognition itself than at classifying and evaluating intelligence. Russell and Norvig's AI Categories provide a task-oriented taxonomy that divides systems into problem-solving, reasoning, learning, and acting (Russell and Norvig 2016). While this classification remains influential, it does not describe a pathway for cognitive progression—that is, the structured development of cognitive features across maturity levels, as represented in Table 4 of the CAIE framework. Francis Chollet's Measure of Intelligence emphasizes skill acquisition and generalization, aligning closely with learning and adaptation (Chollet 2019). However, it underrepresents interactive and collaborative aspects of cognition and does not incorporate optimization or operational efficiency, both of which are increasingly necessary in enterprise-level systems. Finally, Wong's Cognitive Decision Models contribute directly to reasoning and optimization by establishing evaluative criteria for decision-making under uncertainty (Wong et al. 2008a, b). This represents an important conceptual bridge to CAIE, which incorporates optimization as a systemic cognitive dimension and provides a structured pathway for diagnosing both strengths and deficits in complex systems.

Taken together, these models underscore the breadth of approaches to understanding cognition, ranging from biologically inspired reasoning to enterprise-oriented integration. However, from a CAIE perspective, their limitations remain clear: while they excel in highlighting specific dimensions—such as learning, reasoning, or infocommunication—they fall

short in offering holistic tools for assessing maturity, adaptability, and efficiency across multiple domains.

There are several other approaches to evaluating and assessing the adequacy of AI systems, with different objectives: research benchmarking measures technical capabilities, while standards and legal frameworks ensure safety, reliability, and compliance with legal requirements. Table 3 compares four different methods in terms of their objectives, methodology, and legal status.

The three well-known approaches clearly show that the scale of AI assessment ranges from a purely research focus to mandatory legal compliance. Research benchmarks (e.g., Gato) support capability development (ZHYLIN et al. 2025), while the NIST AI RMF (Swaminathan and Danks 2024) provides a framework for risk governance and organizational risk reduction. The EU AI Act, on the other hand, sets out legal obligations that must be met in order to enter the market in the European Union (2024). In addition, the CAIE framework fills a distinct gap by evaluating systems from a cognitive perspective: it maps functions, assesses maturity, and identifies gaps to guide development and structured improvement. Unlike NIST, which emphasizes trustworthiness and risk governance, CAIE is designed as a diagnostic and development-support tool for advancing cognitive and AI systems.

3.2 Identified properties and features of cognitive and artificial intelligence systems

To support systematic evaluation, the CAIE framework identifies 90 core Cognitive and Artificial Intelligence features. These features are organized into six zones corresponding to the categories defined in Table 1 and further detailed in Sect. 4. The zones reflect fundamental capabilities of intelligent behavior, grounded in human cognition studies but extended

Table 3 Comparison of AI-related frameworks, methodologies, and applications

Framework	Nature	Methodology	Goal	Legal status/application
Deep-Mind Gato, AGI Evaluation	Research benchmark	Measurement of performance across multi-modal tasks (text, image, gaming, robotics) within a single model	Examination of generalization ability and AGI potential	Non-mandatory, used in scientific research
NIST AI RMF	Risk management framework	Four core functions: Govern, Map, Measure, Manage; structured risk analysis and measurement	Support organizational governance, trustworthy deployment, and reduction of AI-related risks	Voluntary, but widely recommended for compliance and industry standards
EU AI Act	Regulatory compliance framework	Risk-based classification; verification of data quality, documentation, testing, and human oversight	Ensure safety, protection of fundamental rights, and pre-market conformity	Mandatory in the EU; high-risk systems require certification and CE marking
CAIE	Development quality support	Stage 1: feature-level assessment; Stage 2: GAP analysis; classification based on cognitive levels; development guidance via gap identification	Evaluate cognitive/AI systems by mapping features, diagnosing maturity levels, and identifying gaps to support structured development and improvement	Non-mandatory, used in scientific research and industrial environments

to ensure applicability within artificial systems. This structure provides the foundation for assessing system maturity, adaptability, and contextual performance.

3.2.1 Learning and adaptation

This category focuses on a system's ability to acquire new knowledge, generalize from previous experiences, and adapt to evolving environments and tasks. It encompasses features that measure how efficiently and effectively systems can update their knowledge base and refine their behavior over time.

Core Learning Mechanisms

(1) *General Learning: Fundamental capability of systems to process data and refine behaviors Continuous Improvement through Data Processing and Feedback.*
Learning is the cornerstone of any intelligent system. It refers to the system's ability to continuously refine its processes and improve its performance by processing new data and learning from feedback. This feature allows a system to enhance its understanding of patterns, make better predictions, and evolve its decision-making capabilities. By accumulating experience over time, the system can adjust its internal models to become more accurate and robust, ensuring long-term relevance in dynamic environments. Learning can be assessed by tracking improvements in predictive accuracy, decision outcomes, or task performance over time. Indicators such as the learning rate, reduction in error rates, and the system's ability to generalize knowledge across tasks are key metrics (Mattyasovszky-Philipp and Molnár 2020; Zhang 2019; Mattyasovszky-Philipp et al. 2022).

(2) *Skill Acquisition: Focuses on the rate and effectiveness of new skill acquisition.*
This feature evaluates how quickly and effectively a system can acquire new skills. It focuses on the system's capacity to learn complex tasks, adapt to different learning scenarios, and optimize its learning process over time (Chollet 2019; Button et al. 2008).

(3) *Chunking Learning: Breaks down complex tasks into manageable units to facilitate learning.*
Chunking refers to the process of breaking down complex tasks into smaller, more manageable units, allowing the system to learn incrementally. Systems that can utilize chunking learning demonstrate higher problem-solving efficiency, as they can reuse learned components in different situations (Chollet 2019; Fonollosa et al. 2015; Gobet and Lane 2012).

(4) *Supervised, Unsupervised, and Semi-Supervised Learning: Leverages diverse data structures to enhance adaptability.*
Supervised learning utilizes labeled data to map inputs to outputs, enabling tasks like classification and speech recognition. Unsupervised learning focuses on uncovering hidden patterns in unlabeled data, supporting clustering and exploratory analysis. Semi-supervised learning bridges the gap, combining labeled and unlabeled data to improve model performance where labeled examples are scarce. Together, these methodologies enable systems to adapt to varied data environments, optimizing decision-making and pattern recognition across structured and unstructured datasets (Berry et al. 2019; Moujahid and Dornaika 2025).

(5) *Reinforcement Learning: Learns optimal actions through feedback mechanisms (rewards/punishments).*
This learning methodology enables systems to learn optimal actions through error, receiving rewards or punishments based on the outcomes of their actions. It focuses on adaptive behavior driven by the system's interaction with its environment, optimizing decisions over time (Sun et al. 2008; Duffin et al. 2014).

(6) *Learning with Learning Rate Optimization: Adjusts learning speed to match task complexity and prevent overfitting.*
Learning rate optimization focuses on the system's ability to adjust its learning speed based on the complexity of the task at hand. Systems with efficient learning rate optimization can fine-tune their acquisition process to ensure that they learn rapidly without overfitting or underfitting the available data (Laird 2019; Iiduka 2022).

(7) *Learning trough Mimicking Human Cognition: Includes generative AI capabilities to simulate human-like thought and communication.*
Mimicking human cognition through generative AI, including Natural Language Understanding (NLU) and Natural Language Generation (NLG), enables systems to support human communication by producing content that closely resembles human thought patterns. NLU allows systems to comprehend human language by extracting meaning and intent from text or speech inputs, while NLG helps generate human-like responses. This feature is essential for systems that engage in natural conversations, provide customer service, or perform tasks based on verbal instructions. By simulating human cognitive processes, these systems can produce coherent, contextually appropriate responses, improving human–machine interactions (Mattyasovszky-Philipp et al. 2022; Betti et al. 2023).

Adaptation and Generalization

(8) *Generalization: Applies learned knowledge to new, unseen situations.*
Generalization refers to the system's ability to apply learned knowledge to new and unseen situations, predicting outcomes based on previous experiences. This is a critical indicator of system intelligence, as it measures how well a system can solve unfamiliar problems by recognizing patterns from past interactions (Chollet 2019).

9. *Behaviour Adaptation: Modifies behavior based on environmental feedback and changes.*
Adjusting behavior based on feedback and environmental changes. Adaptation is a system's ability to modify its behavior based on feedback and changes in its environment. This ensures that the system remains effective when faced with evolving external conditions or unexpected disruptions. By adapting, systems become more resilient and responsive, reducing the need for human intervention and increasing operational efficiency. Adaptive systems can handle uncertainties and maintain consistent performance, even when the parameters of their environment change. Adaptation is measured by the system's response time and effectiveness in altering its behavior following feedback or environmental changes. The flexibility of the system to adjust its parameters without degrading performance is another critical metric (Baranyi and Csapo 2024; Mattyasovszky-Philipp et al. 2022; Putnoki 2024; Zimmermann 2024).

(10) *Transfer Learning: Utilizes knowledge from one domain to enhance performance in another domain.*

Transfer learning enables systems to apply knowledge gained in one domain to solve problems in a different, but related domain, enhancing their generalization capabilities. This feature allows systems to leverage past experiences from one context to address new challenges, reducing the need for retraining and improving efficiency in new scenarios. By applying previously acquired knowledge, systems can accelerate learning, adapt to new environments, and perform tasks in untrained domains with greater effectiveness. This concept is inspired by Human-Level AI Systems (HLAI) and frameworks that emphasize transfer learning as a cornerstone for building systems capable of generalizing across domains (Chollet 2019; Zhang 2019; Molina 2022).

(11) *Zero-Shot Learning: Recognizes and categorizes new patterns or situations without prior exposure.*
Zero-shot learning enables a system to recognize, categorize, and apply knowledge to completely new situations or patterns without prior exposure to them. This feature relies on abstract reasoning, where the system uses symbolic descriptions or relationships learned from other contexts to identify new elements. Zero-shot learning is particularly critical in environments where systems encounter new concepts or tasks for which they were not explicitly trained. As highlighted by Francis Chollet's generalization model, zero-shot learning serves as a marker of true system intelligence, demonstrating the ability to generalize knowledge and operate effectively without prior experience (Chollet 2019; Zimmermann 2024; Pourpanah et al. 2022).

(12) *Multi-Domain Intelligence: Operates across various domains applying learned knowledge.*
Multi-domain intelligence refers to a system's capacity to learn and perform tasks across various domains. Such systems can operate in different contexts, applying their learning and decision-making capabilities in a wide range of tasks and environments, making them highly versatile. Multi-domain intelligence is measured by the system's performance consistency across different tasks and domains (Mattyasovszky-Philipp et al. 2022; Putnoki 2024; Rawat 2021).

(13) *Autonomy: Independent Operation and Decision-Making based on pre-learned data and patterns.*
Autonomy represents the system's capacity to operate and make decisions independently, without human intervention. Autonomous systems can assess situations, execute tasks, and even solve problems based on pre-learned data and patterns. This feature reduces human effort and accelerates decision-making processes, making systems more efficient and self-sufficient. Autonomy is assessed by the number of tasks completed without external input, decision-making accuracy in unsupervised conditions, and the system's ability to handle complex tasks independently (Mattyasovszky-Philipp et al. 2022).

Specialized Learning Forms

(14) *Self-Play Learning: Engages in simulated environments to refine strategies and improve autonomously.*
The system autonomously engages in simulated environments (Self -Playing) to improve its performance. This is often seen in strategic systems, where the system continuously refines its skills without external input, iterating through different strategies to optimize outcomes (Silver et al. 2016).

(15) *Cognitive Development through Human–Computer Interaction: Enhances mutual cognitive growth through interactive learning.*
This feature supports the mutual cognitive growth of both the system and its human users through interaction. Systems that foster cognitive development allow users to gain insights, make better decisions, and enhance their knowledge. Such interactions also help systems learn from users. The system's ability to foster learning and improved decision-making in users, as well as its responsiveness to user inputs, can be tracked through surveys and behavioral analysis (Molnár and Mattyasovszky-Philipp 2023).

(16) *Cognitive Exercise, Aging Support, and Digital Skills Development: Supports cognitive skill development across different age groups and genders, ensuring inclusivity.*
Cognitive exercise features are designed to support users of all ages and genders in developing and maintaining cognitive skills. For older adults, these systems help delay cognitive decline by providing mentally stimulating tasks, while younger users benefit from the development of digital literacy and reasoning skills through interaction with digital tools. Ensuring gender equality in these systems means recognizing and addressing any gender-specific needs or differences in how users engage with cognitive exercises and digital development tools, promoting inclusivity and equal opportunities for all users (Baranyi and Csapo 2024; Mattyasovszky-Philipp et al. 2022; Chan et al. 2014).

The Learning and Adaptation category evaluates how systems acquire knowledge, generalize from experiences, and adapt to dynamic environments, forming a foundation for Cognitive Intelligence (Ren et al. 2023). Technologies like Neural Networks (NN), including RNNs and LSTMs, enable pattern recognition and predictive capabilities through supervised and unsupervised learning. Reinforcement Learning (RL), as seen in AlphaZero, fosters adaptive behavior through reward-based optimization, while meta-learning algorithms like MAML enhance learning strategies over time. AutoML platforms, such as Google AutoML and H2O.ai, streamline the learning process for rapid adaptation. Cognitive Information Systems integrate data processing with actionable insights, aligning seamlessly with cognitive frameworks like SOAR, ACT-R, and OpenCog, which emphasize decision cycles and chunking. These technologies and frameworks collectively empower systems to learn, adapt, and tackle real-world complexities with efficiency.

3.2.2 Perception and awareness

This category encompasses a system's ability to sense, interpret, and become aware of both its external environment and its internal states. It measures how well the system processes sensory inputs and maintains situational awareness, critical for systems that interact with real-world environments or complex data streams.

Environmental Interaction, Understanding and Semantics

(1) *Context Awareness: Interprets data within environmental contexts, adapting behavior to real-time scenarios.*
Context awareness refers to a system's ability to interpret data within the scope of its environment or situation, ensuring decisions are relevant to real-time needs. This feature allows systems to analyze incoming data not just in isolation but in the context of

their surroundings or specific scenarios. By understanding the relevance of data based on environmental conditions, systems can prioritize tasks based on urgency or importance, improving decision-making and adaptability in dynamic situations (Mattyasovszky-Philipp and Molnár 2020; Zimmermann 2024; Perera et al. 2014).

(2) *Multimodal and Sensory Integration-Sensor fusion: Integrates and combines data from multiple sensory sources for comprehensive environmental understanding and actionable insights.*
Multimodal perception enables a system to process and integrate data from various sensory sources, such as vision, sound, and touch. This integration creates a comprehensive understanding of the environment, essential for systems operating in dynamic, real-world environments. The combined sensory data provide complementary information that enhances the system's ability to perform complex tasks, such as in robotics. Sensory integration further ensures that information from these different sources is interpreted in a cohesive manner, allowing the system to make informed decisions based on comprehensive environmental data (Baranyi and Csapo 2024; Thatere et al. 2024).

(3) *Visual and Spatial Processing: Processes spatial relationships to navigate and interact within physical spaces.*
Understanding spatial relationships between objects. Visual and spatial processing is essential for systems that need to understand and navigate physical spaces. This feature allows the system to interpret the spatial relationships between objects, making it useful for robotics, drones, and other systems that operate in 3D environments. It enhances the system's ability to avoid obstacles, interact with objects, and plan movements based on a spatial understanding of the environment (Zhang 2019; Molina 2022).

(4) *Perception: General processing of sensory inputs to understand the surrounding environment.* Processing sensory inputs (e.g., vision, sound, touch) to understand the environment. Perception allows a system to gather and interpret sensory data, transforming raw inputs into meaningful information about the environment. This feature is essential for systems that need to navigate or interact with their surroundings, such as autonomous robots or augmented reality applications. By processing inputs like vision, sound, and touch, systems can make informed decisions and execute tasks based on real-time data. Perception can be evaluated by measuring the system's accuracy in recognizing and interpreting sensory data. Metrics include object recognition rates, sensor response times, and the system's ability to correlate sensory inputs with appropriate actions (Molina 2022; Proctor and Proctor 2021).

(5) *Media Multitasking: Handles multiple information streams simultaneously, suitable for dynamic environments with varied inputs.*
Managing multiple tasks or information streams simultaneously. Media multitasking refers to a system's ability to process and respond to multiple streams of information or tasks concurrently. This is particularly useful in environments where various inputs, such as video feeds, audio signals, or sensor data, need to be managed simultaneously. Media multitasking is optional for systems with high-level cognitive requirements but essential in complex, dynamic environments (Zimmermann 2024; Fischer and Plessow 2015).

(6) *Digital Reality: Enhances perception and interaction within virtual/augmented realities, integrating memory with digital environments.*

The ability of a system to simulate and interact with digital environments that are indistinguishable from physical reality. This enhances Perception and Awareness by incorporating systems that can process and act within digital environments as if they were physical ones, increasing the scope of intelligent behavior in simulated spaces (Baranyi and Csapo 2024).

Social and Cognitive Load Management

(7) *Attention Management and Focus: Manages and prioritizes information to help users concentrate on relevant data, reducing cognitive load.*
Attention management enables systems to assist humans by prioritizing critical information and filtering out distractions, ensuring users focus on the most relevant data at the right time. This feature is especially valuable in environments where information overload can reduce human performance, such as in multitasking or decision-heavy tasks. By managing attention and focus, the system helps humans stay productive and make quicker, more accurate decisions. Unlike Inhibition Control and Cognitive Flexibility, which focuses on the system's internal ability to switch tasks, Attention Management and Focus is explicitly user-oriented. It reduces cognitive load on the user by filtering and prioritizing information to assist human attention management, helping users navigate complex or high-pressure situations more effectively (Putnoki 2024).

(8) *Social Cognition: Interprets social cues to improve interactions in environments.*
Social cognition, refers to the mental processes required to interpret social situations. This involves recognizing the emotions, intentions, and thoughts of others, as well as processing signals coming from the social environment. Social intelligence manifests in practice—for example, when someone successfully mediates between two disputing parties. Social cognition, however, operates more in the background—for instance, when someone infers another person's mood from facial expressions and body posture. Thus, social intelligence is the "what you do" in a social situation, while social cognition is the "how you interpret" that situation. The two are closely related, since good social intelligence cannot exist without adequate social cognition. It is particularly important in collaborative environments like healthcare or customer service, where social intelligence can enhance interaction quality (Henschel et al. 2020).

The Perception and Awareness category evaluates a system's ability to process sensory data, maintain situational awareness, and respond effectively to dynamic environments. These features are essential for systems operating in robotics, customer service, healthcare, and autonomous vehicles. Frameworks and architecture models like LIDA, ACT-R, and HRI emphasize sensory integration, attention management, and contextual understanding to enhance intelligent operations. Key technologies in this category include Natural Language Processing (NLP) for understanding text and speech, Computer Vision for image and spatial analysis, and Sensor Fusion algorithms like SLAM (Simultaneous Localization and Mapping) for integrating multiple sensory inputs. Multimodal Perception combines vision, audio, and text data to provide a holistic understanding of environments, while Contextual Computing allows systems to interpret information within specific scenarios. Real-time capabilities are supported by Edge Computing, enabling systems to process sensory inputs with minimal latency. Applications include Voice Recognition tools such as Whisper and Google Speech-to-Text, Optical Character Recognition (OCR) for text extraction. Technologies like IoT Platforms (e.g., Arduino and Raspberry Pi) enable environmental

sensing, while Augmented Reality (AR) Platforms such as ARKit and ARCore facilitate real-time spatial interactions. For robotics, LiDAR Sensors and gesture recognition APIs like MediaPipe enhance spatial awareness and interaction capabilities. These technologies, along with frameworks that prioritize attention mechanisms, enable systems to perceive and respond effectively, ensuring adaptability and functionality in complex and high-pressure environments.

3.2.3 Reasoning and decision-making

This category focuses on the cognitive features that allow systems to analyze data, make decisions, and solve complex problems autonomously. Systems with advanced reasoning and decision-making features are able to simulate human-like thought processes, make logical choices under uncertainty, and even engage in ethical decision-making when required. Reasoning and decision-making features are crucial for systems operating in dynamic, unpredictable environments where they must evaluate multiple factors to make informed decisions.

Planning and Strategy

(1) *Planning: Involves setting strategies and goals based on environmental assessments.*
Planning involves the system's ability to formulate like strategies and set both immediate and future objectives. It requires understanding the current environment, projecting future conditions, and determining the steps needed to reach desired outcomes (Putnoki and Orosz 2023; Berger 2020).

(2) *Executive Functions: Manages and regulates actions to achieve long-term goals.*
Executive functions refer to high-level cognitive processes that allow systems to control and regulate their behavior in pursuit of long-term goals. These functions include organizing, planning, decision-making, and the ability to adjust strategies when encountering new challenges. While Planning focuses specifically on setting goals and strategies, Executive functions ensure that systems can coordinate multiple actions, maintain focus on overarching objectives, and adapt to changing conditions (Cristofori et al. 2019).

Types of Reasoning

(3) *Logical Reasoning and Decision-Making: Uses logic to process information and make decisions.*
Simulating human-like reasoning to evaluate options. Reasoning involves the system's ability to process information logically to make decisions based on evaluating multiple options. This feature mimics human cognitive processes, allowing the system to perform tasks such as comparing outcomes, identifying optimal solutions, and adjusting actions based on the information available. Decision-making enables systems to operate autonomously in environments where human intervention may be minimal, especially in time-critical or resource-constrained situations (Zhang 2019; Chen et al. 2021).

(4) *Probabilistic and Statistical Reasoning: Deals with uncertainty and risk by evaluating probabilities and recognizing patterns.*
Generalization-based reasoning enables systems to make decisions in environments with uncertainty by evaluating the likelihood of various outcomes and recognizing patterns in data. Probabilistic Reasoning quantifies the likelihood of different scenarios, ensuring decisions are guided by statistical probabilities and expected benefits.

Statistical Reasoning supports this process by providing the analytical foundation to interpret data and validate probabilistic evaluations. This process allows systems to adapt and operate effectively in dynamic, uncertain environments, making it essential for robust decision-making under risk or incomplete information (Putnoki and Orosz 2023; Borovcnik 2006).

(5) *Ethical and Moral Reasoning: Applies ethical guidelines and considers moral implications in decision-making processes.*
This feature is crucial for systems operating in sensitive sectors like healthcare, law enforcement, or finance, where decisions have ethical implications and need to align with societal norms. It allows systems to evaluate the potential consequences of actions in a way that aligns with ethical principles and societal values, ensuring decisions are not only effective but also morally sound. Ethical reasoning involves navigating dilemmas by evaluating ethical guidelines, while moral reasoning involves considering the broader societal norms. This plays a critical role in applications where AI systems make life-altering decisions, such as in medical diagnoses or autonomous driving (Kelly and Hamm 2013).

(6) *Dialectical Reasoning: Resolves complex problems by synthesizing opposing perspectives.*
Dialectical reasoning enables systems to analyze and reconcile conflicting viewpoints, promoting balanced decision-making in multifaceted scenarios. This reasoning type involves identifying contradictions, evaluating opposing arguments, and synthesizing them into a unified resolution. By incorporating diverse perspectives, systems can enhance collaborative decision-making, navigate ethical dilemmas, and address ambiguities in complex environments. For example, an AI-powered debate platform might evaluate arguments on both sides of a policy issue to generate an informed recommendation, or an ethical AI system could resolve moral conflicts by weighing contradictory principles. Dialectical reasoning ensures decisions are comprehensive, balanced, and grounded in holistic analysis, making it invaluable for applications requiring nuanced problem-solving (Shiyan et al. 2024).

(7) *Commonsense Reasoning: Applies general knowledge and logic for intuitive decisions in real-world scenarios.*
Commonsense reasoning allows systems to apply general knowledge and logic to make more informed and intuitive decisions. This feature enhances the system's ability to understand real-world scenarios, reason about typical human behaviors, and make decisions that align with expected outcomes. Commonsense reasoning is crucial for systems that interact with unpredictable environments or need to make decisions in novel situations (Molnár and Mattyasovszky-Philipp 2023).

(8) *Analogical Reasoning: Transfers knowledge by drawing parallels between similar situations.*
Analogical reasoning enables systems to recognize similarities between known and unfamiliar scenarios, allowing them to transfer insights or solutions from one domain to another. This reasoning type is valuable for adaptive problem-solving, especially in environments where direct precedents are unavailable. For example, AI might apply lessons learned from managing one type of crisis to address a new, related challenge (Bartha 2013).

(9) *Transductive Reasoning: Infers connections between events based on simultaneous occurrences.*
Transductive reasoning allows systems to identify patterns and make inferences based on observed correlations, even when direct causality is unclear. This reasoning type is particularly useful in early-stage learning or environments with limited data. For example, AI might infer relationships between consumer behaviors and external events to suggest marketing adjustments.

(10) *Critical Thinking: Evaluates arguments and ensures logical coherence in reasoning.*
Critical Thinking enables systems to assess the validity, strength, and relevance of arguments while detecting biases and logical inconsistencies. This feature is essential for systems involved in environments requiring high levels of trust, transparency, and accuracy. By critically analyzing conflicting information and prioritizing relevant data, systems with critical thinking features enhance their decision-making processes (Halpern 2013).

(11) *Fallacious and Circular Reasoning: Identifies and corrects flawed logic to ensure decision-making integrity.*
Fallacious and circular reasoning involves detecting and mitigating errors in logic that can undermine decision-making processes. Fallacies such as overgeneralization, false causation, and circular reasoning (where conclusions are used as premises) can result in invalid or redundant outputs. Addressing these flaws is critical for systems to maintain trust, reliability, and logical consistency. By evaluating logical structures critically, systems can prevent infinite loops, unsupported conclusions, and faulty correlations. For example, an AI analyzing financial risks must avoid reasoning such as 'a company is reliable because it is reliable,' ensuring decisions are substantiated by evidence and sound reasoning principles (Halpern 2013).

(12) *Meta-Reasoning: Evaluates and improves the reasoning process itself.*
Meta-reasoning enables systems to analyze their own reasoning patterns, identify inefficiencies, and adapt strategies for better decision outcomes. This feature is critical for systems engaged in complex, multi-layered decision-making, where iterative refinement can significantly enhance performance (Ackerman and Thompson 2017).

Complex Problem Handling

(13) *Problem Solving: Engages in overcoming complex challenges using various logic.*
Identifying and solving complex problems in uncertain situations. Problem-solving is a system's capacity to address and resolve complex or unfamiliar issues, often in situations where there is no clear or predefined solution. This feature is critical for intelligent systems operating in dynamic environments, such as autonomous systems or AI-driven decision platforms. Problem-solving involves logical reasoning, pattern recognition to overcome obstacles and find the most effective solutions (Molnár and Mattyasovszky-Philipp 2023; Baranyi and Csapo 2024).

(14) *Creativity and Innovation: Generates new solutions and ideas.*
Creativity and innovation features enable systems to support humans by suggesting creative solutions, generating new ideas, or helping with content creation. By autonomously proposing new approaches to problems, systems can assist human users in brainstorming and refining innovative ideas (Putnoki and Orosz 2023; Seidel et al. 2010).

(15) *Decision based on Analytics: Utilizes various types of analytics to derive insights and make decisions.*
Analytics can be divided into five key types: (1) Descriptive Analytics, which focuses on understanding what is happening; (2) Diagnostic Analytics, which explores why something is happening; (3) Predictive Analytics, which forecasts what will happen; (4) Prescriptive Analytics, which recommends actions to take; and (5) Cognitive Analytics, which combines AI and machine learning with traditional analytics to create new insights, leveraging high-performance computing for strategic management decisions and Cognitive Information Systems (Putnoki and Orosz 2023; Sayin et al. 2025).

Human-Centric Synchronization

(16) *Cognitive Entities and Cognitive Resonance: Aligns system reasoning with human reasoning for joint decision-making.*
Cognitive entities and cognitive resonance refer to the system's ability to align its decision-making and reasoning processes with human cognition. Cognitive resonance occurs when human and machine cognition complement and enhance one another, improving decision outcomes through shared insights. This feature supports more efficient and accurate decision-making by leveraging the combined strengths of human intuition and machine processing. Cognitive resonance is closely related to Cognitive Development through Human–Computer Interaction (HCI), as both features involve ongoing alignment and interaction between human and machine cognition, though this feature emphasizes joint decision-making and real-time cognitive synchronization (Mattyasovszky-Philipp et al. 2022; Molnár and Mattyasovszky-Philipp 2023).

(17) *Brain–Computer Interfaces (BCIs): Enhances decision-making through direct brain–system communication.*
Brain–Computer Interfaces (BCIs) allow for direct communication between the human brain and a machine, potentially enhancing decision-making by providing real-time inputs from neural signals. BCIs are especially relevant in areas where real-time human decision-making needs to be augmented, such as in medical applications (Baranyi and Csapo 2024; Mattyasovszky-Philipp et al. 2022).

Behavior and Motivation

(18) *Goal-Oriented Behavior and Motivation: Focuses on achieving specific objectives with an emphasis on task prioritization and motivational influences.*
This behavior involves a system's capacity to prioritize tasks and actions based on long-term objectives or predefined goals. Systems with goal-oriented behavior are designed to pursue specific outcomes, adjusting their actions dynamically as they progress towards achieving those goals. Motivational subsystems drive behavior by influencing the importance of different tasks and managing conflicts between competing objectives (Laird 2019; Sun et al. 2001).

(19) *Emotion-Driven Decision-Making: Integrates emotional aspects into decision-making to enhance human interaction.*
Emotion-driven decision-making refers to the system's ability to incorporate emotional input into its reasoning processes, allowing it to balance rational and emotional influences on its actions. This is particularly important in systems that interact directly with humans, such as customer service applications, where decisions must account for empathy and emotional understanding. Systems with emotion-driven decision-making

can offer more human-like responses and enhance user trust (Samsonovich 2013; Polignano et al. 2021).

The Reasoning and Decision-Making category evaluates how systems analyze data, solve problems, and make decisions autonomously, mimicking human cognitive processes. Cognitive frameworks like SOAR, ACT-R, and CLARION emphasize decision cycles, ethical reasoning, and advanced planning, enabling systems to operate effectively in dynamic, uncertain environments such as healthcare, finance, and autonomous systems. Technologies like Monte Carlo Tree Search (MCTS), as utilized in AlphaZero, for strategic planning, and probabilistic reasoning models like Bayesian networks for handling uncertainty. Commonsense reasoning frameworks, such as OpenCog AtomSpace and Cyc, enhance intuitive and logical inference, while ethical reasoning is supported by explainable AI tools like IBM Watson for Ethics. Brain–Computer Interfaces augment decision-making by integrating real-time neural input, particularly in medical and assistive applications. Cognitive resonance, which aligns human and machine cognition for collaborative decision-making, leverages the strengths of both intuitive human insights and machine precision. Additionally, decision-support systems, constraint solvers, and optimization platforms like Google OR-Tools further contribute to the robustness and adaptability of these systems across diverse scenarios.

3.2.4 Interaction and personalization

This category evaluates how systems engage with humans and other systems, focusing on effective communication, collaboration, and task management. Systems with interaction, personalization and collaboration features can adapt to user needs, adjust their behavior based on input, and work seamlessly alongside other systems or human collaborators. This category also includes features related to NLP, Emotional Intelligence, and task-oriented cooperation.

Emotional and Social Intelligence, Personalization

(1) *Emotional Intelligence: Detects and responds to human emotions to improve interaction quality.*
Emotional Intelligence: Emotional intelligence refers to a system's ability to detect in real-time and respond to human emotions. This is especially valuable in customer service systems where recognizing emotional cues can significantly enhance interaction quality and user satisfaction. By interpreting voice tone, facial expressions, or text sentiment, systems can adjust their responses to be more empathetic and context-aware (Zimmermann 2024).

(2) *Emotional Self-Regulation: Manages the system's emotional responses, maintaining stability in interactions.*
Emotional Self-Regulation: Emotional self-regulation refers to the system's ability to manage its emotional responses in interactions, ensuring stability and consistency, especially in high-pressure or emotionally charged scenarios. Systems with self-regulation can maintain professionalism and avoid inappropriate emotional reactions, which is critical in fields like healthcare or customer service (Molina 2022).

(3) *Social Intelligence: Interprets social cues and context to facilitate appropriate responses in social interactions.*
Social intelligence allows systems to interpret social cues such as body language, tone of voice, or social context, enabling them to respond appropriately in social interactions. Social intelligence is the ability to effectively build relationships, communicate, and cooperate with others. It includes recognizing emotions, selecting appropriate behavior, and managing conflicts. This feature is particularly important in systems designed for collaborative environments, education, or healthcare, where the understanding of group dynamics and interpersonal relationships can enhance interaction quality (Baranyi and Csapo 2024; Mattyasovszky-Philipp et al. 2022).

(4) *Cultural Intelligence (CQ): Adapting Reasoning and Interaction Across Cultural Contexts*
Cultural Intelligence (CQ) refers to the system's capacity to perceive, interpret, and adapt behavior in culturally diverse environments. Unlike general social intelligence, which focuses on reading social cues, CQ includes four distinct but interrelated facets: metacognitive (awareness and regulation of cultural knowledge), cognitive (understanding cultural norms and practices), motivational (interest and perseverance in intercultural situations), and behavioral (the ability to act appropriately in diverse settings). Systems with CQ can dynamically adjust language, tone, reasoning models, ethical frameworks, and interaction protocols to align with the user's cultural expectations and values. This feature is essential for maintaining user trust, improving cross-cultural collaboration, and avoiding communication breakdowns in global or multicultural deployments (Earley and Ang 2003).

(5) *Personalization: Tailors system behavior and responses according to individual user preferences and past interactions. Real-Time.*
Personalization allows systems to adapt their behavior and responses based on the individual preferences and needs of each user. By learning from past interactions, systems can provide tailored recommendations, suggestions, or assistance. It improves user engagement and satisfaction, making the system feel more intuitive and responsive. In addition to typical measures of personalization, behavioral analysis—similar to what is used in Cognitive development through Human–Computer Interaction and personality surveys can be employed to assess how well the system adapts to individual user preferences and behaviors. Personalization can be measured by user satisfaction levels, the system's ability to adapt to preferences over time, and the relevance of personalized recommendations (Mattyasovszky-Philipp et al. 2022).

System robustness and Explainability

(6) *Trust, Transparency, and Explainability (XAI): Enhances understanding of AI decisions, building user trust through transparency.*
Trust and transparency are crucial in building user confidence in AI systems, especially in fields such as healthcare, finance, or legal applications. Explainable AI (XAI) ensures that users understand the reasoning behind the system's decisions, fostering trust and improving user acceptance. Systems that provide transparency in decision-making processes enable users to understand how and why certain decisions are made, which is essential for ethical AI deployment (Wormald et al. 2025).

(7) *Resilience, restore performance after disruptions: Enables recovery and continuity following performance declines.*

Resilient systems are characterized by their capacity to regain normal performance levels after disruptions, as they can effectively recover from declines in performance (Holling 1973; Walker et al. 2004). Lengnick-Hall et al. (2011) describe resilience as the ability of organizations to bounce back from adverse conditions and resume operations as before. Organizations with resilience typically have built-in redundancies and flexibility that aid in restoring original performance levels (Gittell et al. 2006; Mallak 1998; Rudolph and Repenning 2002; Sutcliffe and Vogus 2003; Sheffi 2005).

(8) *Robustness, system resistance: Maintains functional integrity under external stressors.*
Robustness characterizes a system's resistance to external influences (Duchek 2020; Madni et al. 2009; Žiha 2000). It encompasses the system's ability to withstand stressors and maintain performance, demonstrating resilience against adverse conditions (Durach et al. 2015; Haimes 2015). Robust systems typically withstand or absorb stress while preserving their functional integrity (Dubey et al. 2017; Haimes 2009; Kahn et al. 2018; Kamalahmadi and Parast 2016; Sutcliffe and Vogus 2003). Such systems may adjust their operational modes to maintain functionality in the face of rare or unanticipated events that may otherwise lead to failure (Kitano 2004; Carlson and Doyle 2002). In supply chain contexts, robustness implies the capacity to avoid adversity through strategies that include maintaining redundancies and safety stocks (Durach et al. 2015; Gittell et al. 2006; Kobasa 1982; Meyer 1982). However, robustness is limited and beyond a certain point, systems may experience performance degradation (Holling 1973).

(9) *Antifragility, benefit from volatility and uncertainty: Gains strength and adaptability through external stress.*
Systems that deteriorate under uncertainty are fragile, while antifragile systems benefit and grow amidst environments characterized by volatility and unpredictability (Taleb 2012). Taleb's analysis, primarily drawn from financial contexts, describes antifragility through empirical observations, though it is often seen as too abstract for practical application in management (Größler 2020).

Communication and Interaction

(10) *Collaborative Intelligence: Facilitates collaboration between systems and humans to optimize task completion.*
Collaborative intelligence involves systems working alongside humans or other systems to achieve shared goals, automate workflows, or complete complex tasks. This feature leverages human judgment and machine precision, allowing systems to optimize tasks while still relying on human inputs for nuanced decision-making. SOAR general cognitive architecture has been widely used by AI researchers to create intelligent agents and cognitive models of different aspects of human behavior, emphasize the importance of task-oriented collaboration in improving productivity and efficiency in dynamic environments (Laird 2019; Wilson and Daugherty 2018).

(11) *Human–Robot Interaction (HRI): Enables effective interaction between humans and robotic systems, focusing on cooperative task execution.*
HRI focuses on enabling smooth, intuitive interactions between robots and humans. Systems with this feature can interpret social cues, adjust their behavior based on human actions, and work collaboratively with humans in tasks that require physical or cognitive inputs. HRI Frameworks place a significant emphasis on seamless interaction

between humans and robots, particularly in fields like manufacturing, healthcare, and service industries (Galin and Meshcheryakov 2020).

This category evaluates how systems communicate with humans and other systems, adapt their behavior based on input, and collaborate effectively in dynamic environments. It emphasizes NLP, Emotional Intelligence, and task-oriented cooperation, enabling seamless and productive human–machine interactions. Cognitive models like IBM Watson, CLARION, and Human–Robot Interaction (HRI) frameworks underscore the importance of adaptive personalization, transparency, and collaborative intelligence in building trust and efficiency. Systems with these features excel in complex and high-stakes environments such as healthcare, customer service, and industrial automation. Technologies include like Natural Language Generation (NLG), chatbots, and conversational AI platforms such as SAP Conversational AI, which enable real-time, human-like interactions. Emotion-aware systems like Affectiva and Microsoft Azure Face API enhance emotional intelligence, allowing systems to interpret and respond to user emotions effectively. Tools like IBM Emotion AI and Azure Cognitive Services integrate sentiment analysis to improve interaction quality. Explainable AI (XAI) frameworks such as LIME and SHAP ensure decision-making transparency, essential for user trust in applications like healthcare and finance. These technologies and frameworks together empower systems to adapt, collaborate, and interact effectively, transforming dynamic environments into streamlined, efficient ecosystems.

3.2.5 Memory

This category measures a system's ability to store, retrieve, and apply knowledge over time, enhancing decision-making and task execution. Systems with strong memory features can maintain long-term information, allowing them to build a richer understanding of their environments, learn from past experiences, and improve performance based on accumulated knowledge. Memory is crucial for systems operating in dynamic or uncertain environments where decisions are dependent on past interactions.

Types of Memory

(1) *Sensory Memory & DSBM: Sensory Memory, integrated with Data Short Buffer Memory (DSBM), captures and briefly stabilizes raw sensory input to support perceptual awareness and efficient transfer to Short-Term Memory without introducing a separate memory type.* Sensory Memory represents the initial stage of memory processing, responsible for capturing raw sensory input for a brief period ranging from milliseconds to a few seconds. It forms the foundation for perceptual awareness by holding transient impressions of stimuli long enough to allow basic encoding into the system's working memory. In the proposed architecture, Sensory Memory is functionally integrated with the Data Short Buffer Memory (DSBM), which operates as an architectural enhancement. DSBM provides preprocessing and short-term stabilization of sensory input before transferring it to Short-Term Memory. This combination improves responsiveness by ensuring that data is formatted, buffered, and aligned with internal memory models without introducing a standalone memory type (Atkinson and Shiffrin 1968; Baddeley and Hitch 1974; Baddeley 2000; Zhang 2019; Ulrich and Dietz 1985; Wan et al. 2020; Wang et al. 2013).

(2) *Iconic, Echoic, and Haptic Memory: Iconic, Echoic, and Haptic Memory are modality-specific channels within Sensory Memory that briefly store visual, auditory, and tactile inputs to support structured early-stage perception and multimodal integration.* These subcomponents of Sensory Memory represent modality-specific channels that briefly retain environmental input to support early-stage perception. Iconic Memory processes and holds visual information, Echoic Memory handles auditory stimuli, and Haptic Memory stores tactile sensations. Each of these contributes to a coherent sensory experience by providing structured access to different types of environmental stimuli in a format that can be quickly evaluated or transferred to Short-Term Memory. These sensory channels are biologically grounded and are essential in the system's ability to manage and interpret multi-modal information before conscious processing begins (Holding 2014; Sciutti et al. 2019).

(3) *Behavior Action Buffer Memory (ABM): ABM integrates memory-driven cognition into action-ready outputs, synchronizing adaptive system behavior in real-time.* The Behavior Action Buffer Memory is not a traditional memory category but rather a critical control layer that supports the coordination of task execution. ABM receives input from both Short-Term and Long-Term Memory and facilitates the conversion of memory-driven cognition into system-level outputs. It serves as a behavioral interface where retrieved knowledge, contextual inputs, and instructions are integrated and transformed into action-ready representations. By managing task priorities and synchronizing execution across modules, ABM plays a pivotal role in ensuring that the system behaves adaptively in real-time, based on both past experience and current input (Baddeley and Hitch 1974; Baddeley 2000; Wang 2013; Wang et al. 2013).

(4) *Short-Term Memory (Working, Active Memory): Handles temporary information processing, supporting immediate decision-making (DM) and problem-solving.*
Short-term memory allows systems to hold and manipulate information for short durations, enabling real-time responses to dynamic environments. Often referred to as working or active memory, it plays a vital role in fast-paced applications such as natural language processing, real-time analytics, and quick decision-making in critical situations. This memory system typically spans seconds to a few minutes, acting as a gateway between sensory memory and long-term memory storage (Atkinson and Shiffrin 1968; Baddeley and Hitch 1974; Baddeley 2000; Mattyasovszky-Philipp et al. 2022; Zhang 2019; Norris 2017).

(5) *Long-Term Memory Retrieval for Complex Tasks: Retains information over prolonged periods for complex problem-solving.*
Long-term memory allows systems to retain information over extended periods, supporting complex tasks that require a deep understanding of past interactions and accumulated knowledge. Systems with strong long-term memory can recall past experiences, lessons, or decisions and apply them to new situations, allowing for greater flexibility and problem-solving abilities in dynamic environments (Atkinson and Shiffrin 1968; Mattyasovszky-Philipp et al. 2022; Zhang 2019; Norris 2017).

(6) *Advanced Memory, World Model Memory: Stores and updates representations of the environment, aiding in dynamic decision-making.*
Advanced memory systems allow a system to store and retrieve relevant data, improving its ability to learn from past interactions and decisions. World model memory specifically refers to the system's ability to store representations of the environment and

update these representations over time. This feature is critical for systems operating in dynamic environments, as it allows them to keep track of changes, retain context, and make predictions about future states based on accumulated data (Molina 2022; Reale and Russell 2022).

(7) *Procedural Memory (Implicit Knowledge, Skill-Based Knowledge): Holds information about how tasks are performed, facilitating automatic recall and application.*
Procedural memory allows systems to retain learned routines and execute repetitive actions efficiently without relearning. This memory type is particularly valuable in automating processes in robotics and autonomous systems, ensuring consistency and reliability in task execution (Zhang 2019; Cook 2001; Ewen and Mostofsky 2021).

(8) *Motor Skills and Coordination: Remembers and refines physical task execution, essential in robotics and automation.*
Motor skills and coordination are essential in robotics and other systems requiring physical interaction with the environment. Systems that remember how to perform physical tasks based on previous experiences can operate more efficiently and adapt to new tasks by transferring learned motor skills. This ability is crucial in automation, manufacturing, and healthcare, where robotic systems must learn and refine tasks like grasping objects or performing surgeries (Zhang 2019; Pujari 2019).

(9) *Declarative Memory (Explicit Knowledge, Fact-Based Knowledge): Stores and recalls factual data and explicit knowledge, serving as a foundation for reasoning and decision-making processes.*
Declarative memory encompasses both episodic memory (experience-based) and semantic memory (concept-based), but it stands independently as the system's core ability to manage fact-based knowledge. This memory type allows systems to access specific concepts, data, and experiences when needed, enabling logical conclusions and informed decisions. Declarative memory plays a crucial role in intelligent systems, supporting tasks that require accurate retrieval and application of explicit information (Zhang 2019; Bird 2017).

(10) *Episodic Memory (Declarative Memory, Experience-Based): Captures and recalls specific experiences or events, allowing systems to personalize interactions and decision-making.*
Episodic memory supports adaptive systems, enabling them to consider the context of past interactions, recognize patterns, and predict future behaviors based on historical data (Zhang 2019; Dokic 2014).

(11) *Semantic Memory (Declarative Memory, Concept-Based): Structures and retrieves generalized facts, concepts, and knowledge about the world.*
Semantic memory is essential for systems that require reasoning across domains, providing a foundation for knowledge-based processes and decision-making. Knowledge graphs and ontology-based systems are prime examples of technologies leveraging semantic memory (Zhang 2019; Sontheimer 2021).

(12) *Contextual Memory (Including Spatial Memory): Combines spatial and temporal data with situational awareness, enabling systems to adapt recommendations and decisions based on the user's location, time, and activity.*
This memory type retains physical layouts and location-specific data (spatial memory) while integrating broader context to deliver personalized insights. It is essential

for systems like nav igation apps, smart assistants, and recommendation engines that require dynamic adaptation to real-world scenarios(Roediger et al. 2017; Jiang et al. 2005).

Figure 1 illustrates the combined memory map in cognitive systems, grounded in the classical distinction of Sensory Memory, Short-Term (Working) Memory, and Long-Term Memory, as introduced in the modal model of Atkinson and Shiffrin (1968). This foundational tripartite structure was subsequently refined by Baddeley and Hitch through the multicomponent model of working memory, and further extended by Baddeley with the addition of the episodic buffer (Baddeley 2000; Pezzulo 2007). Together, these canonical models provide the conceptual basis for organizing the more specialized memory types in the framework. Sensory Memory, which includes modality-specific channels such as Iconic, Echoic, and Haptic memory (Holding 2014; Sciutti et al. 2019), briefly stabilizes perceptual input and, in the CAIE framework, is enhanced with a Data Short Buffer Memory (DSBM) for preprocessing (Zhang 2019; Ulrich and Dietz 1985; Wan et al. 2020). Short-Term or Working Memory enables temporary information processing and manipulation, supporting real-time reasoning and decision-making (Norris 2017; Mattyasovszky-Philipp et al. 2022). Long-Term Memory encompasses both Explicit (Declarative) knowledge—divided into Episodic and Semantic and Contextual components (Dokic 2014; Sontheimer 2021)—and Implicit (Procedural) knowledge that supports skills, motor coordination, and automated routines (Cook 2001; Pujari 2019). Beyond this classical core, the integrated model extends to include Advanced Memory, World Model Memory, which maintains dynamic representations of the environment to aid predictive reasoning and adaptive decision-making (Reale and Russell 2022). Control-oriented layers such as the Behavior Action Buffer Memory further connect memory-driven cognition with task execution (Wang 2013), ensuring synchronized, context-sensitive system behavior. By embedding these extensions into the Baddeleyan Foundation, the combined map demonstrates how traditional categories interconnect with advanced and specialized forms to support the full spectrum of

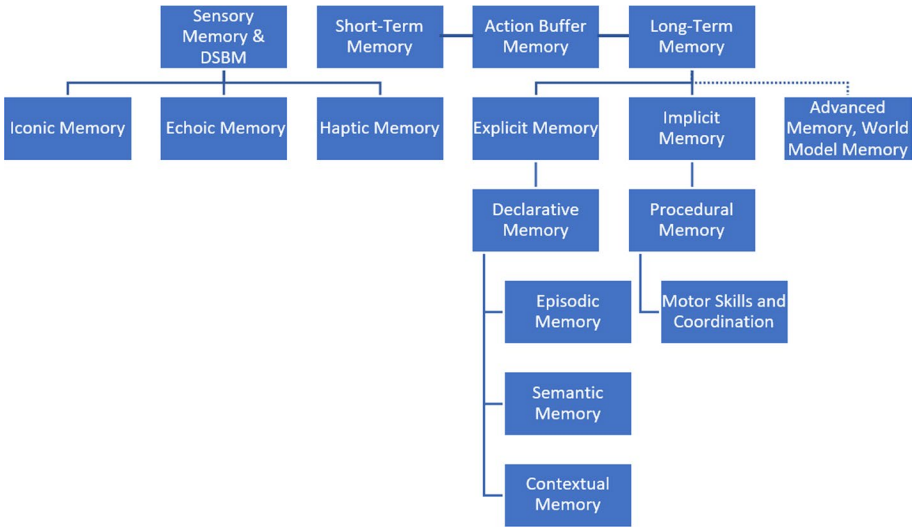

Fig. 1 Combined memory map

cognitive processing. This layered structure highlights the continuity between psychological models of human memory and their application in artificial systems, showing how memory types collectively underpin learning, reasoning, and adaptive intelligence.

Memory support

(13) *Instructive Intelligence and Task Control: Instructive Intelligence transforms memory-driven cognition into prioritized, executable actions, ensuring adaptive task control.*
This feature serves as the critical interface between cognitive memory systems and behavioral execution. It integrates relevant information from both Short-Term and Long-Term Memory to generate prioritized, context-sensitive instructions that guide task execution. By harmonizing recent inputs with accumulated knowledge, the system dynamically selects, refines, and delivers task-relevant directives to the Action Buffer Memory. This enables timely and adaptive system responses aligned with internal cognitive states and external conditions. The integrated mechanism ensures that instructions are not only informed by past experience but also structured for efficient prioritization, allowing the system to handle multiple demands, resolve task conflicts, and maintain goal-directed behavior. As such, it represents a unified control function that transforms memory-driven cognition into organized, executable actions (Wang 2013).

(14) *Memory and Information Retrieval Support: Assists in managing and accessing extensive data efficiently, crucial in knowledge-intensive sectors*.
Systems with memory and information retrieval support act as external memory aids, helping users manage and access vast amounts of information efficiently. This feature integrates tools like online databases, knowledge repositories, and search algorithms to assist users in recalling and organizing information. It is particularly useful in knowledge-intensive industries where rapid and accurate access to data is critical (Baranyi and Csapo 2024; Mattyasovszky-Philipp et al. 2022).

(15) *Knowledge Representation and Organization: Structures and organizes information for efficient storage and retrieval, enhancing reasoning and decision-making.*
Systems with robust knowledge representation features can connect related pieces of information, improving their ability to reason about data and make informed decisions. A key aspect of this feature is Associative Memory, which dynamically links concepts or data points to enhance recall and learning. Associative memory enables systems to identify and utilize relationships between disparate pieces of information, supporting tasks such as pattern recognition, cross-domain reasoning, and advanced decision-making. AtomSpace, from the OpenCog framework, is a prime example of a system capable of organizing vast amounts of knowledge for cross-domain learning and reasoning (Goertzel et al. 2014).

(16) *Forgetting Mechanisms (Selective Memory Optimization): Allows systems to prioritize and retain relevant data while discarding obsolete or redundant information.*
Inspired by biological processes, forgetting mechanisms ensure that memory systems remain efficient and adaptable in dynamic environments by managing data overload. Techniques such as memory pruning, decay functions, or heuristic-based deletion reduce storage burdens while preserving critical information for decision-making and learning. This feature is particularly valuable in systems handling large datasets or operating in evolving scenarios, where the ability to adapt memory dynamically is crucial (Sha et al. 2024).

The Memory category relates to system's ability to store, retrieve, and apply knowledge over time, a cornerstone for improving performance and decision-making capabilities. By integrating both short-term and long-term memory, systems can develop a comprehensive internal representation of their environment, adapt their behavior based on past experiences, and enhance task execution in dynamic or uncertain settings. Cognitive architectures such as ACT-R, SOAR, and OpenCog demonstrate the critical role of declarative, procedural, and long-term memory in enabling intelligent systems to operate effectively across domains. This category encompasses advanced memory systems that store and retrieve relevant data, world model memory for updating environmental representations, and procedural memory for automating repetitive tasks. Declarative memory supports the recall of explicit knowledge, while working memory enables real-time processing for immediate decision-making. Long-term memory retrieval powers the ability to manage complex tasks, drawing from accumulated knowledge and past experiences. Technologies include Neural Networks with Memory, such as RNNs and LSTMs, which are foundational for sequence-based tasks. Memory-Augmented Neural Networks extend this and combining neural networks with external memory storage for more complex reasoning. Knowledge Graphs provide structured representations for cross-domain reasoning, while Cognitive Architectures like SPAUN integrate biologically inspired memory mechanisms. Tools like OpenCog AtomSpace merge symbolic and neural memory for hybrid models. Time-Series Databases like InfluxDB enable efficient storage and recall of time-sensitive data, and Elasticsearch supports context-aware data storage for rapid retrieval.

3.2.6 Optimization and operation efficiency

This category evaluates a system's ability to continuously improve its performance by using feedback from its environment. Optimization involves real-time monitoring and adjustment of internal processes to enhance efficiency, reduce errors, and ensure effective use of resources. Feedback loops play a crucial role in enabling self-correction, allowing systems to learn from their performance and make ongoing adjustments. In addition to operational efficiency, this framework examines the cognitive maturity of a system, focusing on its ability to process information, learn, reason, adapt, and optimize autonomously.

Resource Management

(1) *Computational and Time Efficiency: Focuses on optimizing the use of computational resources and minimizing task completion time.*
This feature evaluates how well the system optimizes computational resources and minimizes time to completion for various tasks. Computational efficiency focuses on reducing processing requirements, while time efficiency measures the latency or speed at which the system can complete tasks and make decisions. Systems that score highly in this feature can perform complex tasks with minimal resource drain and in the shortest time possible (Yang et al. 2014; Mahjoubi et al. 2025).

(2) *Energy Efficiency Optimization: Balances performance with energy consumption to maintain efficiency without excessive resource use.*
Energy efficiency optimization refers to the system's ability to balance its performance with energy or resource consumption. Systems that optimize energy use are particularly valuable in industries where reducing operational costs or environmental impact is a

priority. This feature ensures that systems maintain high performance without consuming excessive resources (Molina 2022).

(3) *Resource Alignment and Scenario Simulation: Aligns resources with strategic objectives and uses simulations to predict and optimize project outcomes.*
Ensure optimal resource allocation and alignment of projects with strategic objectives. Additionally, cognitive technologies, such as AI-powered scenario simulations, analyze and prioritize multiple concurrent initiatives based on their strategic impact, resource usage, and potential return on investment (ROI). Scenario simulations enable the system to predict the outcomes of different project plans and select the most beneficial course of action (Putnoki 2024; Kunc and O'Brien 2017).

(4) *Inhibition Control and Cognitive Flexibility: Balances focus on essential tasks and switches between goals as needed.*
Inhibition refers to a system's ability to filter out irrelevant information, suppress impulsive responses, and maintain focus on essential tasks. Cognitive flexibility complements this by enabling the system to switch efficiently between tasks or goals, adapt to new rules, and handle unexpected changes in priorities. This combination ensures that the system remains focused on priorities while maintaining the adaptability to pivot when necessary, making it versatile and resilient in complex, multitasking environments. Inhibition and flexibility are evaluated through task-switching efficiency, error rates during distractions, and the system's ability to reprioritize tasks based on dynamic conditions. Performance improvements in managing competing tasks or stimuli also serve as critical indicators of these features (Cristofori et al. 2019).

(5) *Scalability: Manages increased complexity or workload without performance loss.*
The ability of a system to expand and manage increased workloads or complexity without compromising performance or efficiency. Scalable systems can handle growing demands, adapt to the addition of new resources, or respond to increased data processing requirements (Zhang 2019; Putnoki 2024).

Feedback and Adaptation Mechanisms

(6) *Optimization Through Feedback Loops: Uses real-time environmental feedback to continuously refine performance.*
Feedback loops are essential for systems that need to optimize their performance continuously. These systems rely on real-time input from their environment to make immediate adjustments and improvements. By monitoring performance data, feedback loops ensure that the system operates as efficiently as possible, constantly refining its processes based on current conditions (Krishnamoorthy and Skogestad 2022).

(7) *Self-Monitoring: Allows systems to autonomously track and adjust their performance in real time.*
Self-monitoring refers to the system's ability to autonomously track its own performance and make real-time adjustments. This is crucial for maintaining operational effectiveness, especially in environments where manual intervention may not be feasible. Self-monitoring systems are designed to detect inefficiencies or issues and correct them immediately without external input (Putnoki 2024; Chiba and Krichmar 2020).

(8) *Dynamic Model Adaptation: Adapts models dynamically to maintain alignment with operational changes and strategic planning.*
Ensure the seamless integration of different systems while dynamically updating models (AS-IS, TO-BE) in real-time. These tools reduce downtime, enhance synergy during

mergers or technological upgrades, and ensure that enterprise models remain aligned with current operations and future planning (Putnoki 2024).

(9) *Meta-Learning (Learning to Learn): Enhances learning strategies based on past experience to adapt more effectively to new challenges.*
Meta-learning refers to the system's ability to improve its learning strategies based on past experiences. Rather than just learning new tasks, systems with meta-learning features enhance their efficiency by refining how they learn, enabling faster and more effective adaptation to new challenges. This feature is essential for systems that encounter a wide variety of tasks or operate in unpredictable environments (Zhou et al. 2018; Khoee et al. 2024).

(10) *Metacognition (Self-Monitoring and Evaluating Performance): Involves high-level self-assessment and strategic improvement planning.*
Metacognition enables a system to evaluate its own performance at a strategic level, beyond immediate corrections. It involves self-awareness and reflection, allowing systems to assess their overall effectiveness and make long-term, strategic improvements. This is especially important in systems designed for complex environments where high-level decision-making and long-term planning are required (Putnoki 2024; Zhang 2019).

Security and Risk Management

(11) *Security Efficiency: Ensures system integrity and protection against external threats.*
Security refers to the system's ability to protect itself from external threats, such as adversarial attacks, data breaches, and unauthorized access. This feature ensures the system's integrity, confidentiality, and reliability by employing security protocols that defend against malicious inputs or interference. Security measures are especially important in environments where AI systems process sensitive data or operate autonomously (Uddin et al. 2025).

(12) *Risk Efficiency: Balances novelty-seeking and operational safety, crucial for systems in unpredictable or high-risk environments.*
Risk efficiency focuses on the system's ability to learn and operate in uncertain or potentially dangerous environments while minimizing exposure to risk. Systems that demonstrate high risk efficiency can engage in safe exploration and learning, balancing novelty-seeking behavior with operational safety. This feature ensures that systems continue to learn and adapt in new, unpredictable environments, without exposing themselves to undue risk. The ability to balance exploration of new situations (generalization) with the mitigation of hazards is critical for systems operating in real-time or high-risk environments such as autonomous vehicles, robotics, or healthcare.

(13) *Governance and Compliance Monitoring: Automates compliance checks and governance to ensure adherence to regulations and standards.*
Governance and compliance monitoring ensures that systems adhere to internal and external standards, such as legal regulations or organizational policies. This feature automates the process of monitoring compliance and streamlines governance workflows. It is especially relevant in industries where strict adherence to regulations is required, such as finance, healthcare, or manufacturing (Putnoki 2024; Baranyi and Csapo 2024; Putnoki and Orosz 2023).

(14) *Audit readiness, Documentation automation: Automates the creation and updating of documents, increasing efficiency and reducing errors in documentation processes to ensure audit readiness.*
Documentation automation refers to systems that can autonomously generate, update, and corporate documentation. By automating this process, systems reduce the risk of human error, ensure consistency in documentation, and significantly lower the manual effort required for record-keeping. This feature is particularly valuable in large organizations where maintaining up-to-date architecture documentation is crucial for regulatory compliance and operational efficiency (Putnoki 2024).

General Performance and Strategy Optimization

(15) *Performance Evaluation: Assesses effectiveness based on specific metrics and adjusts operations to optimize outcomes.*
Performance evaluation involves assessing the overall effectiveness of a system in achieving its goals. Systems with this feature can measure their success based on predefined metrics, such as accuracy, efficiency, or user satisfaction, and adjust their actions accordingly. Performance evaluation helps systems fine-tune their operations and optimize outcomes in future scenarios (Gudelli 2023).

(16) *Cognitive Control and Strategy Adjustment: Manages cognitive processes and recalibrates strategies based on environmental changes or performance data.*
Cognitive control refers to a system's ability to regulate its cognitive processes, including focusing attention, suppressing distractions, and adjusting strategies when encountering new challenges. Strategy adjustment involves recalibrating decision-making frameworks based on changing environmental conditions or performance outcomes. Together, these features ensure that systems remain agile and adaptable in the face of uncertainty.

The Optimization and Operation Efficiency category evaluates how systems refine performance through real-time monitoring, feedback mechanisms, and adaptive strategies. By incorporating self-monitoring and metacognitive features, these systems continuously improve efficiency, reduce errors, and align with strategic objectives. Optimization is crucial for systems operating in dynamic, high-stakes environments where adaptability and resource efficiency are paramount. Cognitive architectures like SOAR, CLARION, and OpenCog, alongside Russell and Norvig's AI models, underscore the importance of feedback loops, resource alignment, and risk efficiency in fostering operational excellence. Technologies in this category include Machine Learning (ML) and continuous optimization algorithms for enhancing system responsiveness and scalability. Self-monitoring tools like AWS CloudWatch and Microsoft Azure Monitor provide operational insights, while dynamic adaptation frameworks like Kubernetes enable auto-scaling and load-balancing. Process automation technologies, such as Robotic Process Automation (RPA) and Intelligent Process Automation (IPA), enhance efficiency by automating repetitive tasks and integrating advanced AI for complex processes. Business Process Management (BPM) and Digital Process Automation (DPA) streamline and optimize end-to-end workflows, ensuring adaptability across industries. Energy optimization algorithms from Green AI initiatives prioritize sustainable operations, while intelligent frameworks for performance evaluation and cognitive control enhance decision-making under changing conditions. Together, these tools

and frameworks position systems to evolve autonomously, remain resilient, and achieve long-term operational excellence.

4 Cognitive and artificial intelligence evaluation framework zones

Grounded in the theoretical foundations presented in Sect. 3, the CAIE framework provides a structured, domain-independent approach for evaluating the Cognitive and Artificial Intelligence features of complex systems. It is designed to reflect not only the presence and performance of individual features but also their role in system-wide maturity. The evaluation is organized into six core dimensions, each corresponding to a distinct cognitive capability:

(A) *Learning and Adaptation* assesses the ability of systems to acquire, generalize, and adapt knowledge over time.
(B) *Perception and Awareness* measures the interpretation of input signals and the maintenance of contextual or environmental awareness.
(C) *Reasoning and Decision-Making* evaluates logical inference, strategic planning, and autonomous decision execution.
(D) *Interaction and Personalization* analyzes human–machine interaction, user-tailored responses, and communication effectiveness.
(E) *Memory* examines how systems store, retrieve, and utilize both short-term and long-term knowledge.
(F) *Optimization and Operational Efficiency* measures system optimization through feedback loops, self-evaluation, and resource efficiency.

Unlike domain-specific models that focus on isolated aspects of intelligence, CAIE offers a holistic view that incorporates learning, perception, interaction, optimization, and memory mechanisms. This ensures that systems are assessed not only by their static capabilities but also by their potential to grow and adapt in complex environments.

4.1 Motivation and framework rationale

While existing intelligence evaluation frameworks—such as those developed by Russell and Norvig, Chollet, and SOAR—have contributed valuable perspectives to the field of Artificial Intelligence, they tend to focus on narrow domains, single-task capabilities, or generalized theoretical modeling. As a result, they often lack the mechanisms to evaluate real-world systems in a modular, scalable, and domain-adaptive manner.

These models typically fall short in several critical areas: they are not designed to assess specific features or implementation-level details within diverse system types; they do not link low-level functionalities to system-wide cognitive maturity; and they rarely provide a unified method for evaluating both developmental feasibility and structural readiness. Moreover, many existing models overlook the importance of adaptability—one of the most fundamental characteristics of intelligent behavior.

To address these limitations, the Cognitive and Artificial Intelligence Evaluation (CAIE) framework introduces a two-stage methodology that is both structured and adaptable:

- *Cross-Domain Versatility* the framework is applicable to a wide range of systems, from enterprise architectures and embedded AI modules to cognitive robotics and simulation platforms.
- *Feature-Level Modularity* each cognitive feature is evaluated independently through structured scoring, allowing system architects to assess fine-grained performance while still maintaining alignment with broader functional categories.
- *Adaptable Use and Benchmarking* the framework supports both internal system diagnostics and external comparison. Stage 1 alone enables practical benchmarking across systems without requiring universal expectations, while Stage 2 supports system-specific planning and prioritization.
- *User-Weighted Adaptability* the evaluation process incorporates user-defined importance ratings, enabling context-sensitive scoring that reflects domain-specific priorities and preferences without compromising structural comparability.
- *Strategic Maturity Analysis* in Stage 2, the framework supports dependency-aware evaluation by identifying development gaps, premature activations, and cognitive bottlenecks. This facilitates roadmap creation and design feasibility checks.
- *Pipeline-Based Validation* each feature is aligned with system workflows through an input–processing–output analysis, ensuring conceptual grounding while verifying applicability across practical implementation scenarios.

To ensure completeness, a pipeline-based discovery and validation method was applied. This examined how features interact across full system workflows—tracing information flow from perception through memory access, reasoning, and final output generation. Each feature was included only if it could be mapped to at least one realistic scenario or enterprise use case, ensuring both relevance and non-redundancy. As a result, the CAIE framework provides not just a static taxonomy but a dynamic model of cognitive integration that scales with system complexity and purpose.

5 Cognitive and artificial intelligence evaluation framework levels

The CAIE framework defines five abstract maturity levels to structure the progression of Cognitive and Artificial Intelligence features. These levels are not tied to any specific technology but describe the increasing integration and complexity of the features.

Each level introduces more autonomy, adaptability, and generalization, allowing evaluators to determine whether a feature is absent, reactive, analytical, integrative, or self-improving. This provides a modular and scalable reference structure.

- *Level 1—Very Low* absent or fixed-rule. Systems operate based on predefined logic without adaptation, learning, or dynamic input processing.
- *Level 2—Low* basic and reactive behavior. Feedback is minimally used for local adjustments, but systems lack deeper contextual understanding or predictive capabilities.
- *Level 3—Medium* predictive and analytical behavior. Systems anticipate events or optimize performance based on structured data, using historical or probabilistic logic.
- *Level 4—High* adaptive and integrated reasoning. Systems integrate contextual, multimodal, or ethical dimensions into their reasoning, exhibiting dynamic decision-making.

- *Level 5—Very High* self-improving and generalized intelligence. Systems continuously learn and improve, transfer knowledge across domains, and dynamically reorganize internal processes.

Table 4 outlines how each of the six core dimensions progresses across these five levels, based on standardized functionality focus.

5.1 Mapping cognitive features to maturity levels

Philipp Matrix as Table 5 provides a theoretical mapping of cognitive features to maturity levels across the six core zones. It outlines which features typically appear at earlier stages of system development and which are more advanced, offering a structured perspective on cognitive progression. Philipp's work on cognitive modeling (Molnár and Mattyasovszky-Philipp 2023) serves as the conceptual basis for this stage-based feature mapping. The rows of the matrix represent the six core zones of the CAIE framework from (A) Learning and Adaptation to (F) Optimization and Operational Efficiency, defined in Sect. 4. The columns of the matrix correspond to the five maturity levels (from Level 1—Very Low to Level

Table 4 System-Zoned progression of features across Maturity Levels

	Level 1	Level 2	Level 3	Level 4	Level 5
A	No learning or adaptation	Basic feedback-based learning	Predictive generalization and skill acquisition	Context-aware autonomous learning	Full self-optimization and cross-domain adaptation
B	No signal interpretation	Basic input signal processing	Context-aware perception and prioritization	Multi-modal coordination and integration	Full situational tracking and predictive awareness
C	No reasoning or planning	Rule-based responses	Analytical reasoning and foresight	Ethical and contextual problem-solving	Autonomous, collaborative decision-making
D	No interaction or personalization	Reactive and static interaction	Context-adaptive dialogue and input matching	Personalized, real-time engagement	Fully collaborative and emotionally-aware interaction
E	No memory	Short-term data buffering	Historical recall and predictive use	Memory-context integration and reflection	Self-reinforcing memory with lifelong learning
F	No optimization	Basic loop-based feedback	Structured optimization and refinement	Self-monitoring and contextual adaptation	Dynamic self-optimization and continuous improvement

Table 5 Philipp Matrix—mapping of features to maturity levels

	Level 1	Level 2	Level 3	Level 4	Level 5
A	None	A4, A11	A1, A2, A3, A6, A8	A5, A7, A10, A15	A9, A12, A13, A14, A16
B	None	B1, B8	B2, B3	B4, B6, B7	B5
C	None	C1	C2, C3, C6, C14, C15, C18	C5, C7, C9, C11, C19	C4, C8, C10, C12, C13, C16, C17
D	None	None	D2, D3, D4, D5	D1, D6, D7, D8	D9, D10, D11
E	None	None	E1, E2, E3, E4, E5, E7, E14	E9, E8, E12, E13, E15	E6, E10, E11, E16
F	None	F1	F2, F4, F15	F5, F6, F7, F9, F16	F3, F8, F10, F11, F12, F13, F14

5—Very High), reflecting how each cognitive capability manifests and develops in progressively more complex systems, defined in Sect. 5. Each cell of the matrix specifies the expected maturity expression of a feature within a given zone and level. For example, in the (A) Learning and Adaptation zone, A1 (General Learning) is positioned at Level 3, where systems begin to demonstrate predictive generalization and skill acquisition. In contrast, higher-order functions such as A13 (Autonomy) and A14 (Self-Play Learning) appear only at Level 5, corresponding to full self-optimization and cross-domain adaptation. By providing this structured layout, the Philipp Matrix not only classifies which functions belong to which maturity stage, but also allows cross-referencing to the detailed function descriptions presented in Sect. 3. This mapping helps identify areas where foundational features are well established, as well as regions where higher-level features are sparse—potentially revealing underdeveloped or underexplored domains.

In addition to highlighting feature density across maturity levels, the matrix supports strategic planning. It allows system designers to better understand which features form the basis for more complex capabilities and where future development efforts may be focused. Within Stage 2 of the CAIE framework, Philipp Matrix can optionally serve as a baseline reference for assigning expected maturity levels (L_i^{expected}), especially in cases where domain-specific targets are not available. While it does not define system-specific dependencies and Dependency Matrix, it offers a coherent developmental trajectory that can help validate whether implemented features align with broader cognitive design principles. In general, the matrix acts as both a diagnostic lens and a roadmap tool, supporting architectural completeness evaluations and helping prioritize intelligent system evolution.

6 Cognitive and artificial intelligence evaluation framework model

Building upon the theoretical foundation and feature mappings outlined in the previous sections, the CAIE framework transitions from a structural taxonomy into a practical evaluation model. This model assesses both the presence and maturity of cognitive features,

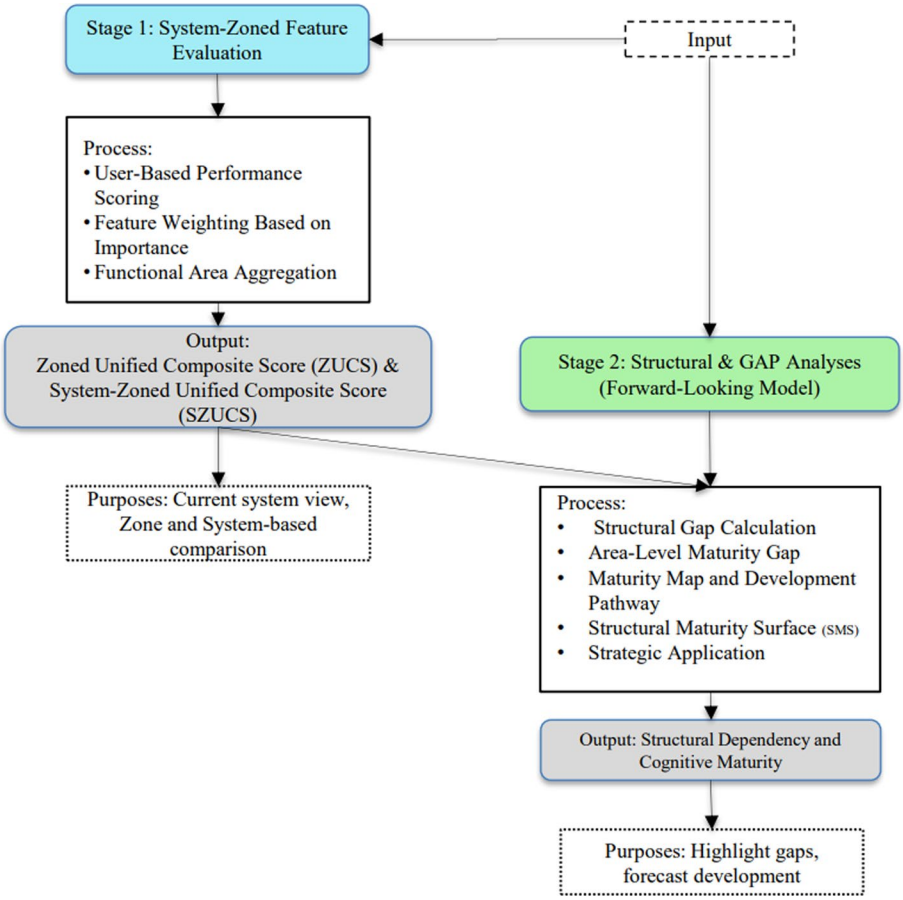

Fig. 2 Stages 1 and 2 of CAIE

integrating system architecture, user experience, and strategic expectations. The evaluation unfolds in two stages:

- Stage 1 focuses on feature-level analysis, evaluating implemented features in terms of system maturity, user-perceived performance, and contextual relevance. This stage also enables zoning and system-based comparisons.
- Stage 2 introduces a structural perspective, identifying maturity gaps, inter-feature dependencies, and architectural development pathways, supported by their visualization.

Figure 2 presents the two-stage structure of the proposed evaluation framework. Stage 1 (System-Zoned Feature Evaluation) builds on solid epistemological and cognitive science foundations, drawing from established theories and architectures to provide a diagnostically reliable baseline of the system's current state. In contrast, Stage 2 (Structural and GAP Analyses) is intentionally introduced as a forward-looking theoretical model. While it does not yet rest on empirical validation, its role is to extend the framework beyond static assess-

ment by forecasting structural dependencies and cognitive maturity trajectories. This prospective orientation ensures that Stage 2 functions as a conceptual pathway for anticipating system development, with its final adequacy to be confirmed through subsequent validation and practical case studies.

To support this analysis, the model incorporates targeted surveys across three key roles: developers, users and stakeholders, ensuring a comprehensive and multi-source data foundation for both diagnostic and strategic evaluations.

6.1 Survey roles and data inputs

To accurately evaluate both the presence and relevance of cognitive features within a system, the CAIE model relies on structured input from the three distinct roles. Each role answers a different set of questions aligned with their perspective:

- Developers/System Architects
  - Feature Presence (P_i): "Is Feature *i* implemented in the system?" Binary value: $P_i \in \{0, 1\}$
  - System-Level Maturity (L_i^{system}): "What maturity level (L1–L5) does Feature *i* operate at currently?"
- Users
  - Performance Score (V_i): "How well does the system perform Feature *i*?" Scale: 1 = Poor, 10 = Excellent
  - Importance Rating (w_i^{user}): "How important is Feature *i* to your experience?" Scale: 1 = Not Important, 10 = Very Important
- Stakeholders (Optional)
  - Expected Maturity (L_i^{expected}): "What target maturity (L1–L5) should Feature *i* reach?"
  - Strategic Weight ($w_i^{\text{stakeholder}}$): "How critical is Feature *i* to strategic success?" Scale: 1 = Low, 10 = Mission-critical

This role-based structure ensures traceable, multi-source evaluation inputs, enabling both present-state assessment and strategic gap analysis in later stages. While the listed questions are structured for surveys targeting developers, system architects, or stakeholders, the corresponding data may also be sourced from architectural documentation, technical specifications, or system design documentation where available.

6.2 Stage 1: system-zoned feature evaluation

This stage provides a quantitative evaluation of each cognitive feature independently. A top-down methodology is employed, in which system architects first define the implemented features, and user experience is then used to assess performance and contextual relevance.

6.2.1 User-based performance scoring

Implemented features are evaluated by users in terms of actual system performance and experienced quality. The normalized performance score of each feature is calculated as:

$$f_i = P_i \cdot \frac{V_i}{V_{\max}}, \tag{1}$$

where f_i is a final normalized score of feature *i*; $P_i \in \{0, 1\}$ means feature presence, provided by architects; V_i is a user-assigned performance score (Likert scale, e.g., 1–10); $V_{\max}$ is a maximum scale value (typically 10).

The purpose of this formula is to normalize the performance evaluation to a [0, 1] scale while ensuring that features not implemented in the system do not contribute to the final score.

6.2.2 Feature weighting based on importance

User-provided importance ratings are used to assign weights to features, reflecting their contextual significance:

$$w_i = \frac{1}{k} \sum_{1}^{k} R_{ik}, \tag{2}$$

where w_i is the importance weight of feature *i*; R_{ik} is an importance rating given by user *k* for feature *i*, where $R_{ik} \in \{1, 10\}$; *k* is the number of user responses.

This weight allows the aggregation to reflect not only how well a feature performs, but also how powerful and important it is in user perception. The presence of a function in a system is the prerequisite for the user to be able to evaluate it at all. This is necessary because the usefulness of the function, and thereby its performance, directly influences the overall efficiency of the system. However, this effect can only fully manifest if the user's subjective assessment also supports the function's usability at the highest possible level.

6.2.3 Functional area aggregation

Each cognitive feature is associated with a broader functional area *F*. The overall area score is calculated as a weighted average of the performance of features within that area:

$$S = \sum_{i \in F} w_i \cdot f_i, \tag{3}$$

where *S* is an aggregate score for area *F*; w_i is an importance weight of feature *i*; f_i is a normalized score of feature *i*.

The purpose of this calculation is to generate a score for a cognitive functional area, incorporating both system performance and user relevance, yielding a more comprehensive area-level evaluation.

6.2.4 System-zoned unified composite score

The whole system's cognitive profile across all functional areas—six dimensions, which is expressed as *Z*. An additional unified score can be computed to represent an overall system-level performance as $ZUCS_i$. This score serves as a Functional Area or dimension F_i normalized indicator, and the *SZUCS* formula presents the whole system-wide composite score.

$$Z = \{F_A, F_B, F_C, F_D, F_E, F_F\}, \tag{4}$$

$$ZUCS_i = \frac{S_i}{w_{i_max}}, \tag{5}$$

$$\text{SZUCS} = \frac{\sum S_i}{\sum w_{i_max}} = \sum \left(\text{ZUCS}_i \cdot \frac{w_{i_max}}{\sum w_{i_max}} \right), \tag{6}$$

where S_i is a functional area score for dimension *i*, where $i \in \{A, B, C, D, E, F\}$ and $F_A \cup F_B \cup F_C \cup F_D \cup F_E \cup F_F = F$.

The *SZUCS* score serves as a normalized, aggregate indicator of system-wide cognitive capability. It enables benchmarking across solutions, supports high-level decision-making, and consolidates multi-dimensional evaluations into a single interpretable metric.

6.3 Stage 2: structural dependency and cognitive maturity

While the first stage offers an independent assessment of each cognitive feature, the second stage introduces structural modeling. This layer captures the hierarchical dependencies among cognitive features and their relative maturity, thereby enabling a deeper analysis of cognitive architecture, readiness, and integrity.

6.3.1 Structural gap calculation

A dual-layered maturity model is adopted. Each feature F_i is characterized by:

- A realized system-level maturity (L_i^{system}), indicating the current operational state.
- An expected maturity level (L_i^{expected}), reflecting strategic intent or ideal structural placement.

The introduction of L_i^{expected} enables context-aware benchmarking. These expectations are not universal; rather, they are defined by stakeholders, architects, or domain experts to suit the unique goals of the system. For instance, meta-learning may be essential in an adaptive AI platform but irrelevant in a logistics optimizer. The structural maturity gap for each feature is calculated as:

$$\Delta l_i = L_i^{\text{expected}} - L_i^{\text{system}}, \tag{7}$$

The structural maturity gap for each feature can be classified into three categories, and

$$\Delta l_i = \begin{cases} \text{Feature has been activated before prerequisites were satisfied,} & \text{if } \Delta l_i < 0, \\ \text{Feature maturity aligns with expectations,} & \text{if } \Delta l_i = 0, \\ \text{Feature is underdeveloped relative to its target,} & \text{if } \Delta l_i > 0. \end{cases}$$

Δl_i quantifies the alignment between the current feature and the strategic vision. It acts as a driver for feature-level prioritization.

6.3.2 Area-level maturity gap

To evaluate maturity alignment at the functional area level, the average maturity gap is computed as follows:

$$g = \frac{\sum \Delta l_i}{|F_j|}, \tag{8}$$

where g is an average maturity gap for zone F_j; F_j is a set of features in system zone j; $|F_j|$ is a number of features in zone j.

A domain-level insight provided by g, enabling cross-functional comparison and area-specific strategy formulation.

6.3.3 Maturity map and development pathway

The full set of Δl_i and g values can be visualized using:

- *Heatmaps* for spatial maturity visualization.
- *3D Terrain Models* where the vertical axis reflects development priority.
- *Dependency Matrices* revealing upstream/downstream relationships and violations.

Purpose: Data visualization tools are used to visually represent unstructured or structured data so that patterns, trends, and relationships within the data can be more easily recognized and interpreted.

The numerical presentation of research results is useful as it quantitatively demonstrates outcomes and correlations, but graphical visualization plays an indispensable role in contemporary data analysis. For the graphical representation of the Δl_i and g values, effective tools are available, which will be applied in the next phase of the research. As visualization techniques, the applicability of Heatmaps, 3D Terrain Models, and Dependency Matrices has been considered, and the detailed development of these visualization approaches will be carried out in subsequent work.

6.3.4 Structural maturity surface (SMS)

To support roadmap generation, a Structural Maturity Surface (SMS) is constructed by plotting each feature against:

- Its maturity level (L_i^{system}).
- Its maturity gap (Δl_i).
- Its weighted importance (w_i).

The following markers are identified on the SMS:

- *Local minima* $\Delta l_i > 0$ and $L_i^{\text{system}} \leq 2$ with high $w_i \rightarrow$ critical underdeveloped fea-

tures.

- *Global maxima* $L_i^{\text{system}} \geq 4$ and $w_i > 6 \rightarrow$ strategic strengths.
- *Premature activations* if F_i depends on F_j but $L_i^{\text{system}} > L_j^{\text{system}}$.
- *Missing key features* $P_i = 0$ and $w_i > 6 \rightarrow$ significant implementation gaps.

Purpose: The SMS turns abstract gap data into actionable patterns, helping teams spot bottlenecks and formulate evidence-based development strategies.

6.3.5 Strategic application

Based on SMS analytics, the following actions are supported:

- *Development targeting* features with large $\Delta l_i \cdot w_i$ scores are prioritized.
- *Roadmap construction* dependencies and foundation-first logic inform sequencing.
- *Investment optimization* budget can be aligned with areas offering the highest cognitive returns.
- *Goal alignment* strategic intent (via L_i^{expected}) can be validated or adjusted based on system readiness.

Purpose: The structural layer complements Stage 1 by evaluating not only 'how much' intelligence is present, but 'how well it is organized'. It acts as the architectural conscience of the system—ensuring that intelligence is built on solid, interdependent layers rather than isolated features.

The second stage transforms the CAIE framework from a feature-level snapshot into a multidimensional model of cognitive alignment. Whether used in its full or partial form (without expected values), this stage introduces depth, dependency awareness, and strategic direction into AI feature evaluation.

7 SAP BTP use-case evaluation

To validate the CAIE framework, a pilot evaluation was conducted on SAP BTP, a modular and AI-enabled enterprise platform for data management, automation, and application development. The analysis was based on user-only feedback from 21 respondents, all active BTP users. The resulting system-wide intelligence profile (*SZUCS* score) was 0.244, based on the following area-level *ZUCS* values:

(A) Learning and Adaptation: 0.395,
(B) Perception and Awareness: 0.089,
(C) Reasoning and Decision-Making: 0.127,
(D) Interaction and Personalization: 0.261,
(E) Memory: 0.171,
(F) Optimization and Operational Efficiency: 0.366.

Despite being based solely on user feedback, the results offer a surprisingly clear picture of cognitive strengths and gaps in SAP BTP. Dimensions like (A) and (F) score well, thanks

to features such as General Learning, Behavior Adaptation, Governance, and Self-Monitoring—showing that SAP BTP already supports adaptive performance and operational intelligence.

However, critical gaps were revealed in (B) and (C): users consistently rated these features as important, but reported they were largely absent. This misalignment highlights an opportunity for SAP to expand its cognitive roadmap with higher-level perception, social context-awareness, and advanced decision-making features.

The CAIE framework was positively validated through both structural justification and empirical support. For RQ1 and RQ2, 95.24% of participants affirmed that cognitive and AI features can be objectively evaluated and meaningfully adapted to user-defined relevance. Additionally, 90.48% confirmed that cognitive capability can serve as a valid core metric (RQ3) because it plays an important role in recognizing when more advanced Cognitive Information Systems may be needed. Several users remarked that the framework effectively captures long-missing cognitive features that leadership frequently highlights as gaps. Additionally, participants appreciated the clarity and structure of the framework itself, noting that the features included were highly relevant and well-chosen.

In summary, this pilot demonstrates that even limited-scope feedback can reveal actionable insights when structured through CAIE. A full implementation—with input from SAP BTP system architects, expected maturity levels, and functional dependencies—would not only complete the picture but could also directly support SAP's future cognitive development and product strategy.

8 Results

The Cognitive and Artificial Intelligence Evaluation (CAIE) framework has been successfully formalized as a two-stage methodology for evaluating intelligent-like behavior of AI-based systems through their cognitive properties, in a structured, scalable, and domain-independent manner. A functional taxonomy comprising over 90 cognitive features has been defined, systematically categorized into six core dimensions: Learning and Adaptation, Perception and Awareness, Reasoning and Decision-Making, Interaction and Personalization, Memory, and Optimization and Operational Efficiency. Each feature has been evaluated for conceptual completeness, architectural traceability, and real-world applicability through pipeline-based modeling.

A unified five-level maturity scale has been established to represent the progression of each feature from rule-based implementation to generalized and self-improving intelligence. The maturity of these features has been theoretically mapped using the Philipp Matrix, which has enabled a structured alignment between expected developmental stages and actual implementation levels. This mapping has supported the identification of maturity asymmetries, underrepresented domains, and premature functional activation risks.

Within Stage 1, a multi-input scoring structure has been constructed, aggregating implementation status's binary value (P_i), normalized performance feedback (f_i), and user-defined importance weights (w_i). These components have been combined to generate per-feature intelligence scores. Aggregation across functional zones has yielded Zoned Unified Composite Scores (*ZUCS*), forming the cognitive performance profile of any system. A system-wide composite score—System-Zoned Unified Composite Score (*SZUCS*)—has

been introduced as the final outcome of Stage 1, capturing weighted cognitive expressiveness while preserving feature-level resolution.

Stage 2 has been defined as a structural analysis layer, incorporating architectural dependencies, maturity gaps, and roadmap alignment. Although full structural data have not been collected, the model has been formally equipped with the logic for dependency matrices, Structural Maturity Surfaces (SMS), and strategic prioritization maps. As a result, the CAIE framework has been positioned to serve not only as a measurement tool but also as a design and optimization methodology.

PRISMA-based literature review was conducted, covering six cognitive dimensions across five major academic databases from 2020 to 2024. The review confirmed that certain areas—such as optimization—were heavily saturated in existing literature, while others—such as memory—was underrepresented. This justified the multidimensional balance implemented in the CAIE framework.

9 Discussion

The results of this research highlight that Cognitive and Artificial Intelligence systems can be evaluated through a structured and theoretically grounded framework that supports objectivity, user adaptability, and cognition-centric analysis. The CAIE framework enables a level of granularity previously unattainable in the evaluation of AI capabilities and cognitive functions by decomposing intelligence into discrete, mappable features and providing a domain-agnostic architecture.

A notable strength of the CAIE structure lies in its flexibility. Features can be independently assessed and interpreted according to both technical implementation and subjective user relevance. This is particularly valuable in environments where systems serve diverse roles and stakeholders, and where traditional metrics such as computational speed or task automation may not sufficiently capture the adequacy of cognitive features and adaptive decision support.

To support the methodological validity of the Cognitive and Artificial Intelligence Evaluation framework, three key research questions (RQs) were posed. These questions collectively address the potential for structured, adaptive, and cognitively grounded evaluation of AI systems. Below, each question is answered with detailed justification based on theoretical foundations, practical implementation logic, and survey-based feedback.

- RQ-1: Is it possible to compare Cognitive and AI systems in a way that provides an objective measure of applicability and usability?
  The findings confirm that the CAIE framework provides a quantifiable and structured model that bridges the gap between theoretical architecture and system-level deployment. It enables comparison by translating diverse system features into normalized/standardized scores based on feature presence, user experience, and functional maturity. This is not only theoretically grounded—as seen in frameworks like SOAR, ACT-R, and Chollet's intelligence measure—but also aligned with practical enterprise goals through metrics like the Zoned Unified Composite Score and the overall *SZUCS* index. The modular, system-zoned structure allows comparing systems on a like-for-like basis even if they differ in domain or complexity. Additionally, the Philipp Matrix acts as a cogni-

tive scaffolding, showing expected maturity paths and highlighting gaps, allowing systems to be benchmarked against theoretical trajectories. The presence of both functional and architectural evaluation levels ensures that usability is not defined in isolation but in relation to context, priority, and interaction.

- RQ-2: Can (Cognitive and AI systems related) measurability be ensured in an adaptive way by developing a rating system that incorporates user preferences?
  The CAIE framework proves to be inherently adaptive by integrating real-world user feedback and role-based evaluation metrics. Performance (V_i) and importance weights (w_i) are entirely user-driven and reflect actual contextual usage patterns, not abstract assumptions. This supports dynamic benchmarking that evolves with user behavior or strategic shifts. The system is adaptable over time, based on feedback loops, across roles such as developers, users, and stakeholders, and across maturity levels through Δl_i gap analysis. The architecture also allows future expansions, for instance integrating learning rates or sensor-based feedback, without altering the core structure. Furthermore, the model supports cross-user aggregation while preserving individual variability, a key requirement for truly adaptive AI evaluation.
- RQ-3: Can cognitive capability levels of AI systems be treated as a key scoring factor, revealing the need for Cognitive Information Systems (CIS)?
  Results support that cognitive capability can be operationalized as a core scoring factor. By explicitly mapping features to maturity levels (L1–L5) and tracking gaps (Δl_i), the framework quantifies cognitive sophistication. A lack of key features in high-impact zones, such as reasoning, memory, or feedback adaptation, becomes a signal not just of underperformance but of structural misalignment, often indicating that a CIS would be beneficial or even required. The Structural Maturity Surface (SMS) enables strategic insight by identifying local weaknesses, global strengths, and premature activations that expose architectural flaws. Moreover, features with high user weight but low system presence—or where Δl_i is significantly greater than zero—reveal precise areas where CIS capabilities could unlock latent value. The framework thus transforms cognitive features into a measurable, actionable parameter, positioning CIS not just as an add-on but as a systemic need when certain thresholds are not met.

Additional theoretical justification reinforces the legitimacy of this approach and the survey results were evaluated significantly positively: RQ-1 and RQ-2 received near-unanimous agreement, while RQ-3 also gained strong support. Respondents emphasized that leadership often raises concerns about the lack of such cognitive features in enterprise systems, and the CAIE framework was praised for finally capturing what had long been missing.

A key limitation that constrained the full activation of Stage 2 was the absence of architecture-level input, which in turn prevented the derivation of outputs such as maturity gap metrics, feature-dependency mapping, and roadmap diagnostics. While the Philipp Matrix already serves as a theoretical scaffold for cognitive maturity, its empirical application remains restricted without baseline reference values and structural dependencies. These limitations not only define the boundaries of the present study but also point to clear directions for future work, where a complete dependency matrix and its graphical representation will be developed and validated within more structurally complete evaluation environments.

10 Conclusion

This paper introduced the CAIE framework as a structured, scalable, and cognitively grounded model for evaluating artificial systems. By unifying feature-level assessments with user-centered prioritization, and by designing for both implementation-based benchmarking and structure-aware maturity analysis, the framework advances the field of intelligence evaluation.

Key contributions include the integration into a structured framework of over 90 cognitive features, the introduction of the new Zoned Unified Composite Scores (*ZUCS*), and the derivation of the System-Zoned Unified Composite Score (*SZUCS*). The framework maintains a strong theoretical foundation by applying structured maturity references, such as the Philipp Matrix, while allowing for contextual adaptability through user-defined weights. Notably, the CAIE framework supports comprehensive evaluation across all architectural elements, encompassing both Cognitive and Artificial Intelligence capabilities. While such features are central to the framework, it remains inclusive of non-AI systems. These can be evaluated within the same structure, as the scoring model accounts for all features—AI-based or not—by assessing their maturity, relevance, and presence through a unified methodology.

Future research could focus on the full activation of Stage 2, enabling structural maturity mapping, dependency modeling, and roadmap formulation. A major area of expansion involves the integration of a cognitive recommendation engine, which would generate system-specific suggestions for feature development, prioritization, or architectural adjustment based on maturity progression. Additional research directions include the extension of the Philipp Matrix for domain-specific maturity models and the automation of P_i detection via system documentation parsing.

In summary, CAIE establishes a new paradigm for system evaluation, unifying structural coherence with functional depth. Its precision, adaptability, and forward compatibility position it as a foundational instrument for academic research and for the evolution of both cognitive and traditional information systems.

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Declarations

Conflict of interest The authors declare no conflict of interest.

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